

Evaluating Land Use Change *from a Pre-disaster Recovery Planning Perspective*: A Model to Analyze Post-disaster Settlement Location

(災害事前復興計画の観点からの土地利用変化の評価：災害後の定住地分析モデル)

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Table of answers to reviewer's questions and comments

Reviewer	Questions/Comments	Answer	Pages
Ubaura Sensei	What did you assume?	When a disaster occurs, a local government is faced with the challenge of redeveloping devastated areas. So, I assumed that learning from post-disaster experiences of the past in other countries could help develop future recovery plans. To do so, I thought that a growing population is vital to regenerate an area that has been affected by a disaster. I also thought that population growth is associated with the physical environment which supports people's activities. Then, I thought that people's activities are best reflected in land cover. Therefore, if population trends and land cover changes can be analyzed based on past post-disaster experiences, perhaps local governments can use the results as indicators for envisioning recovery plans for newly affected areas.	10-14
Ubaura Sensei	What did you prove?	Chapter 3 proved that vehicular accessibility and built-up density surrounded by farmlands played an important role in the stability of population growth areas. Chapter 4 proved that resettlement sites caused minor land use growth effects (15-18% of the district's total). These findings can be used for the land use microsimulation model in Chapter 5 . Then, making the model, I saw that a land use network generated urban growth along the network increasing transit and access to livelihood options, thus reducing travel expenses. This result is one example for only this case based on the model's performance, but I think this method gives some information to consider the image of future affected areas.	10-14
Ubaura Sensei	Why can others use your model?	In this study, I applied lessons from the Great East Japan Earthquake and the Merapi volcano eruptions to the model and reached a conclusion. Admittedly, this conclusion only refers to two cases. However, this research result was obtained by monitoring and evaluating a 10-year recovery and there has never been any research like this before. I expect to investigate other cases in the future and hopefully improve the model's accuracy. I believe that by accumulating this kind of research and evolving it into a tool that contributes to the formulation of recovery plans, other local governments will be able to use it.	14
Ubaura Sensei	Why did you use population data for a depopulating region?	First, I used population data to determine whether an area was growing or struggling to grow. Given that most areas were found to struggle with growth, I thought there was no need to consider land cover as in the case of Indonesia.	10
Ubaura Sensei	Why did you use land cover data for urbanizing region?	Urbanizing regions experience a higher intensity of land use growth than depopulating regions and the maximum effects of resettlement were required to avoid future land cover degradation.	12
Ubaura Sensei	Can you use your model in depopulating regions?	Future application of this model in a depopulating region is possible using Chapter 3 and Chapter 4's result in Chapter 5's method as input data.	14
Ubaura Sensei	Chapter relationships are unclear.	Pages 10-14 were included to explain chapter relationships, I also added a figure and explanation at the beginning and end of Chapters 3-5. Also, a summary of chapter relationships is provided in p.55.	10-14, 23, 32-33, 42-43, 55.
Ubaura Sensei	Where does 63% come from in your research?	Thank you for the observation. I included an additional page explaining how this value is obtained.	51
Koshimura Sensei	Improve model figures to show relations between Chapters 3 and 4 with 5.	Thank you for the comment. I added a new figure which explains the reasons and relations between chapters as well as a new figure and summary of chapter relationships in page 55.	10-14, 55
Koshimura Sensei	How does your model handle uncertainty from past resettlement cases?	I included the following statement to explain it: "Past resettlement cases can bring uncertainty. To handle this uncertainty, this study suggests that only local roads able to sustain more than the expected land use growth are surveyed as a safety measure to avoid land use dispersion".	57
Koshimura Sensei	Presentation is too wordy.	I have shortened the amount of text as much as possible without eliminating answers to the questions/comments provided during the reviews. I also increased figure sizes and exchanged wordy sentences for bullet statements.	5-57
Kubota Sensei	How is your chapter 3 result new?	Basically, I got two results: 1. Regional Characteristics obtained by the analysis (p.31). 2. Using the result, I could proposed a unique method for resettlement area detection (p.32).	31-32
Kubota Sensei	Explain about population movement in rural areas and its relation with your model.	In areas where the population declined even before the Great East Japan Earthquake, there was a tendency for the population to decrease even further during the reconstruction process and beyond. From this trend and other results from the analysis, I proposed Method 1: Resettlement area detection near regional cities in depopulating regions on p.32. I also added a limitation in p.56 where I explain how resettlement near regional cities could cause population movement to their peri-urban areas.	56

Contents

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Chapter 2. Post-Disaster Recovery Research and Development

Chapter 3. Evaluating Urban Growth Structures for Resettlement Area Detection after the 2011 Great East Japan Earthquake

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Chapter 5. Land Use Microsimulation Model for Livelihood Diversification after the 2010 Merapi Volcano Eruptions

Chapter 6. Conclusions

Chapter 1. Post-Disaster Resettlement and Recovery Problem

- Rapid urbanization in the periphery of urban areas poses an increasingly urgent call for planners in the context of natural disaster risk.
- After a major disaster occurs, disaster-affected areas are likely to experience land use and population movement due to relocation, as well as mobility issues related to livelihoods.
- However, post-disaster resettlement in urbanizing rural areas represents a unique opportunity to manage urban growth, while shrinking cities offer a chance to explore non-traditional approaches to generate growth ([King and Gurtner, 2021](#)).
- Relocating communities can consume a lot of resources and time. This is sometimes not in accordance with the expectation of people who want to get back to their previous state.
- Many past examples of resettlement have “failed” to relocate communities in places that offer livelihood choices, allow population growth, and ensure sustainable ecosystems ([Miller, 2018](#); [Contreras et al., 2013](#); [Tanaka, 2023](#); [Sina et al., 2019](#)).
- Successful cases overcome this disadvantage through the availability of reliable infrastructure, mobility options, and proximity to previous and new livelihood options. These cases also restore damaged ecosystems and maintain the ecological balance of resettlement areas.
- Failed cases have not been able to sustain previous livelihoods and promote diverse livelihood options for the socioeconomic improvement of relocated people. Some of these cases have not been able to conserve the environment leading to land degradation and contamination ([Ishihara and Tadono, 2017](#); [Ingram et al., 2006](#)).

- Pre-disaster recovery plans as seen in Figure 1 can provide a strong foundation for post-disaster reconstruction projects.
- However, there is always a need to address pre-existing issues.
- Encouraging a town or city to resolve these issues post-disaster, may prove too challenging.
- Therefore, an organized planning process that is ready to be implemented before a disaster can ensure that local needs and opportunities to increase local resilience are considered ([FEMA, 2017](#)).

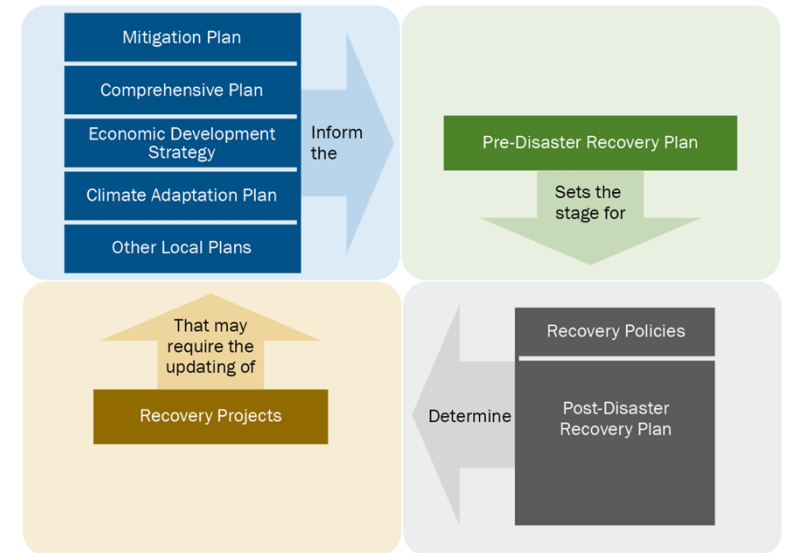


Figure 1. Disaster recovery planning cycle.
Source: FEMA, 2017

In the case of the 2010 Merapi volcano eruptions, a disaster resettlement initiative relocated thousands of households in the Cangkringan sub-district of Yogyakarta, Indonesia, shown in Figure 2 ([The World Bank, 2016](#)).

The affected livelihoods were based on dairy farming and agriculture, while tourism was the secondary employment ([Rindrasih, 2018](#)).

Households of the largest resettlement site, Pagerjurang, currently believe that resettling has granted them safety from the volcano.

However, safety did not translate into secured livelihoods ([Miller, 2018](#)):

1. Even though tourism became a main source of livelihood diversification, it did not promote choices of livelihood in ample spheres of life;
2. Middle-aged farmers and the elderly are too tired of commuting to the dairy farming cooperative and their land on Mt. Merapi;
3. The Cangkringan sub-district is undergoing rural urbanization which is shifting land-use structures that generate place-attachment.
4. Therefore, rural identities are at stake, risking tourist attractive value ([Umay et al., 2020](#)).

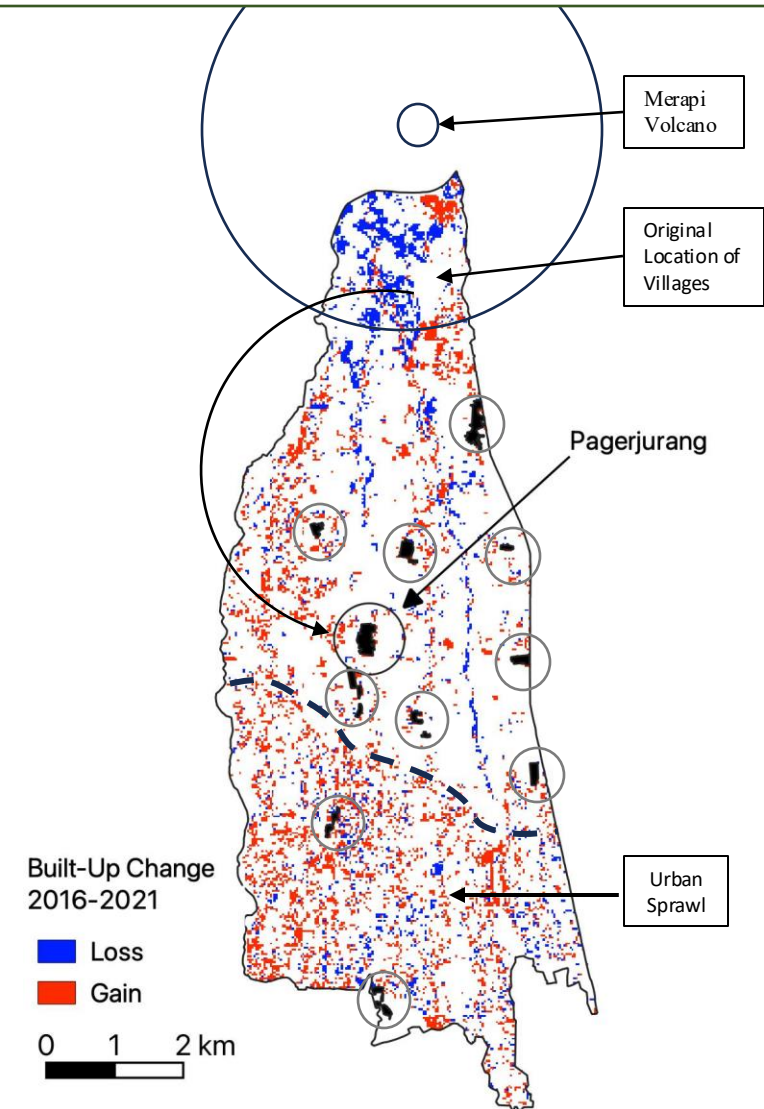


Figure. 2. The Cangkringan Sub-District of Yogyakarta, Indonesia.



Figure 3. A newly constructed residential area above the tsunami danger zone in Higashi-Matsushima, Miyagi Prefecture.

Source: <https://www.nippon.com/en/japan-topics/g01048/>



Figure 4. Rokugo district, Wakabayashi Ward. Sendai City projects.

Source: https://sendai-resilience.jp/en/efforts/government/development/inland_relocation.html

- In contrast to resettlement in an urbanizing context, the 2011 Great East Japan Earthquake resulted in the first major disaster that Japan has faced since the era of population decline (Tanaka, 2023).
- The reconstruction and recovery of housing and communities was undertaken at the core of all recovery plans.
- However, collective relocation projects have not been able to move residents “collectively” to safer areas (Tanaka, 2023).
- Only a third of the victims rebuilt their homes using disaster reconstruction support programs (e.g., Figure 3 and Figure 4).
- Also, families moved away due to restoration delays (Muroi, 2022), community disintegration (Abe, 2014), and inconvenient life in temporary shelters (Miyasada and Maly, 2021).
- Moreover, many relocation projects resulted in (Tanaka, 2023):
 1. the peripheral expansion of residential areas;
 2. the formation of low-density urban areas; and
 3. the creation of a decentralized regional structure based on the separation of jobs and residences.

Research Question

Is it possible to derive an appropriate reconstruction and recovery plan to resolve these issues from prior data (population, land cover) and surveys of relocating residents' travel routines?

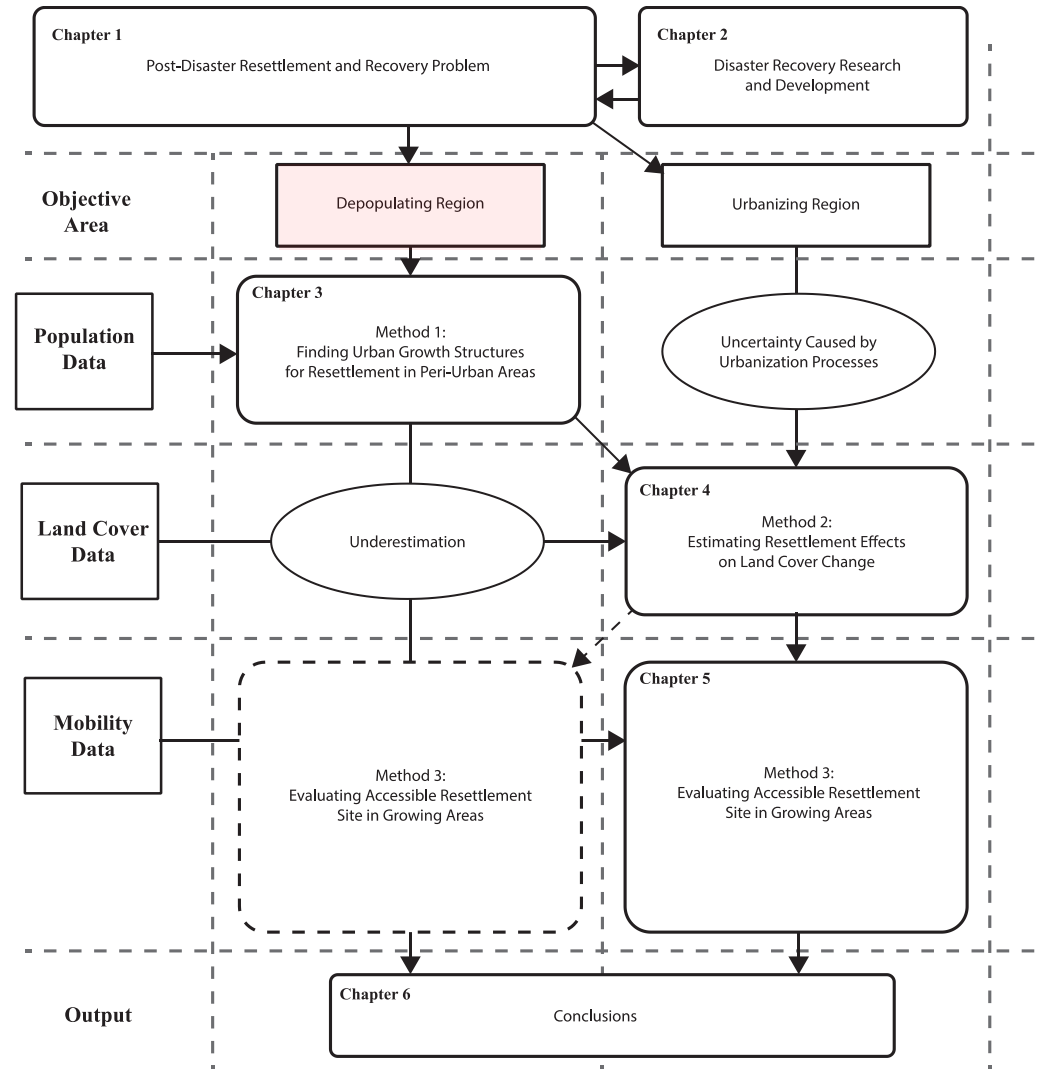
Research Purpose

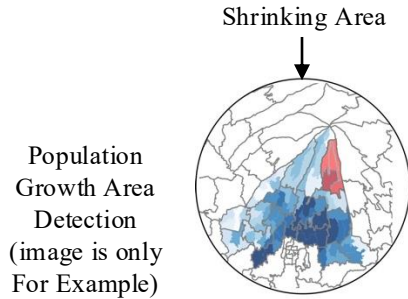
The purpose of this research is to clarify the relationship between preliminary population, land cover data and resettlement location, as well as mobility and livelihoods, and to develop a model that will contribute to the formulation of future recovery plans.

The proposed model is separated into three methods described in Chapter 3, Chapter 4, and Chapter 5.

Chapter 3 selected the 2011 East Japan earthquake to explore population growth in a depopulating region for two reasons:

1. First, I used population data to determine whether an area was growing or struggling to grow.
2. Since most areas were found to struggle with growth, I thought that there was no need to consider land cover as in the case of Indonesia.



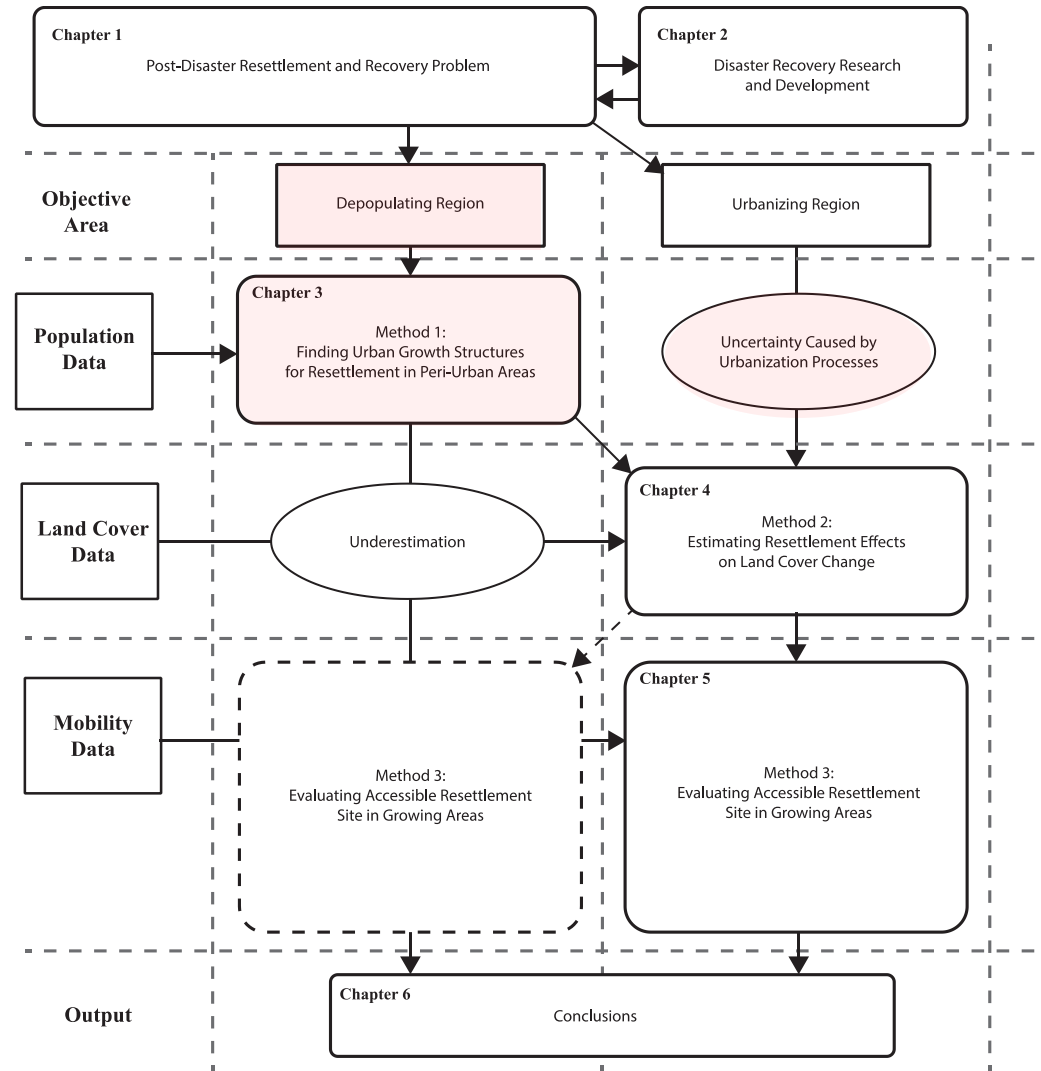


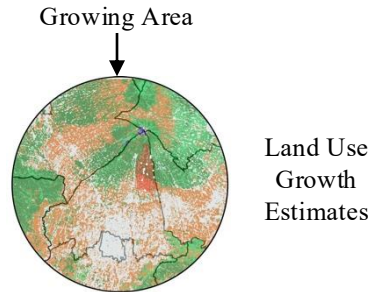
Purpose: This chapter evaluates the relationship between post-disaster population growth and urban structures to identify where future resettlement could undergo population growth.

Results: Growing population areas were analyzed.

- (i) Pre-disaster trends were found unchanged.
- (ii) Vehicular accessibility and built-up density surrounded by farmlands play an important role in the stability of population growth areas.

Findings: Suggest that pre-disaster population data can be used to locate future resettlement areas likely to undergo urban growth.





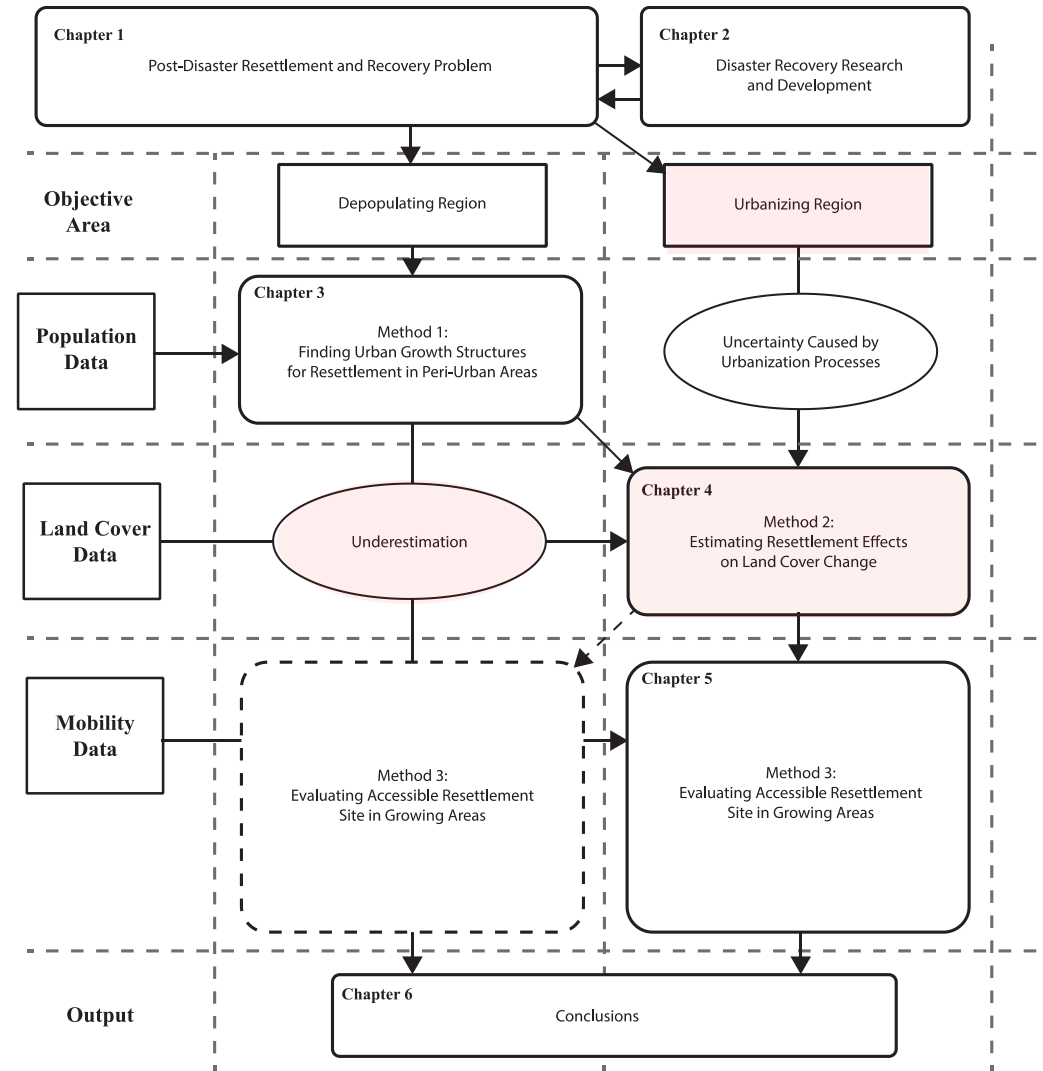
Chapter 4 selects Merapi affected areas to estimate land cover change because urbanizing regions experience a higher intensity of land use growth than depopulating regions and the maximum effects of resettlement were required to avoid future land cover degradation (Creutzig et al., 2019; Ishihara and Tadono, 2017; Harada et al., 2015).

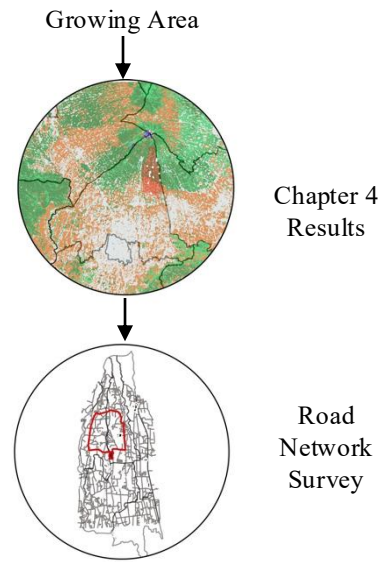
Purpose: To evaluate land change interactions prompted by post-disaster resettlement sites.

Hypothesis: Urban resettlement in rural areas has a significant effect on land use change.

Results: Proved that resettlement sites caused minor land use growth effects (15-18% of the district's total).

Findings: Indicate no change in trends.





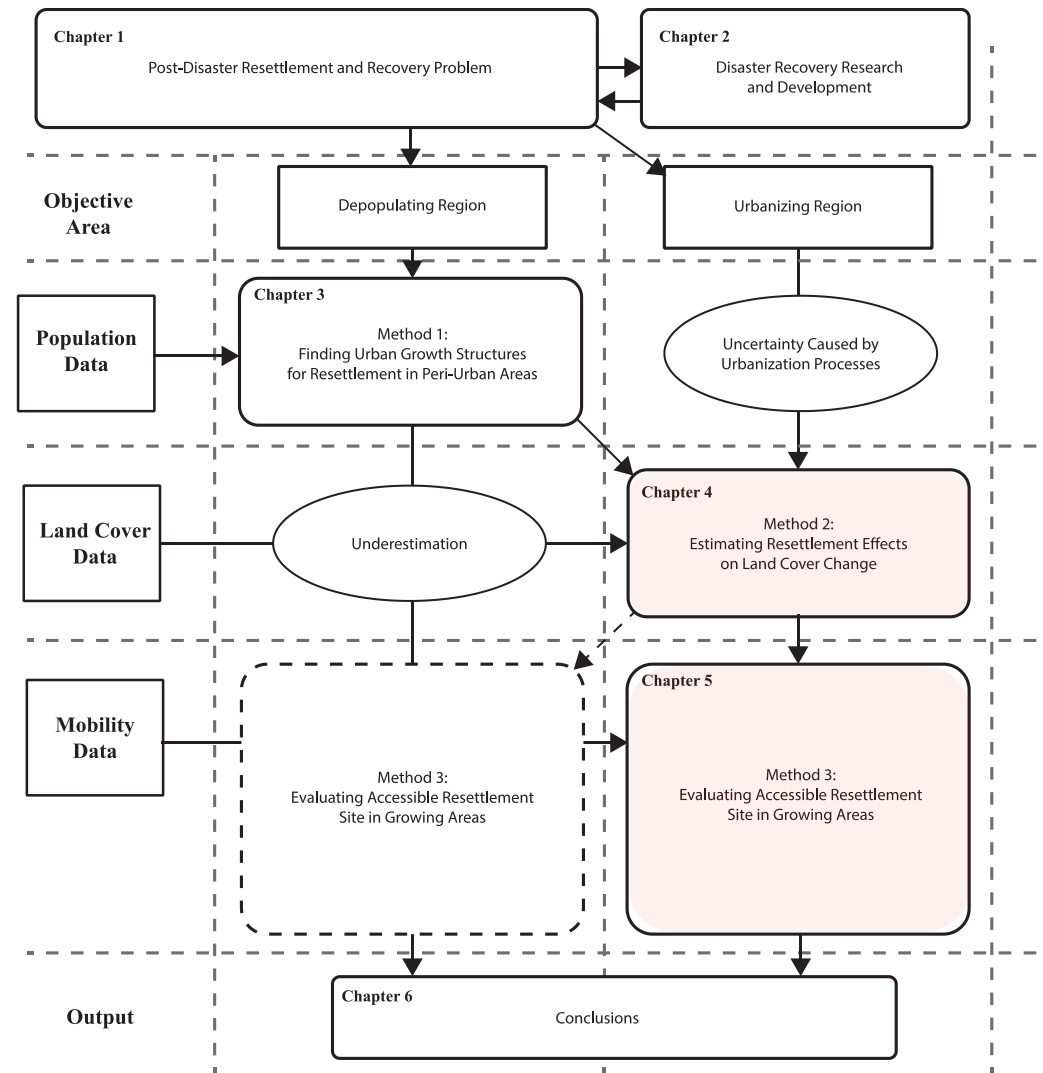
Chapter 4's findings were used to find a candidate resettlement site for evaluation.

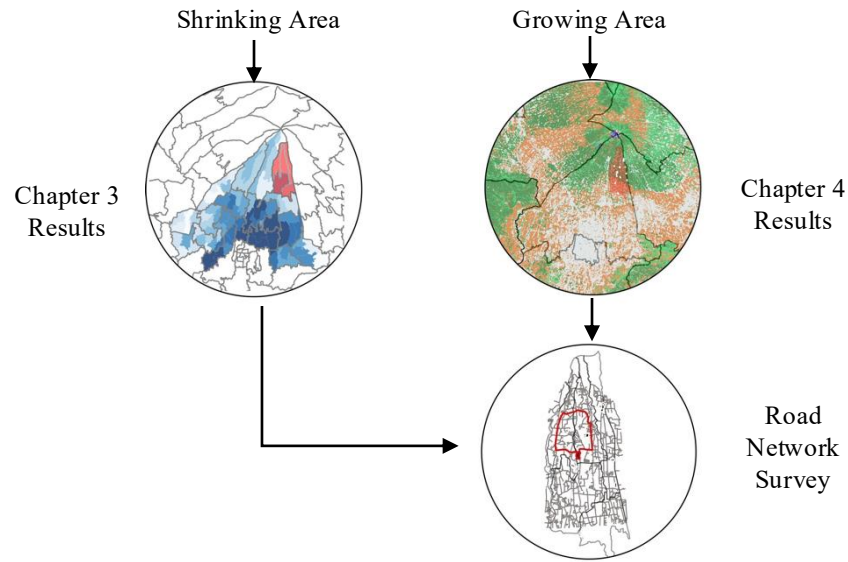
Purpose: Chapter 5 evaluated a settlement's location in Merapi affected areas to clarify whether rural livelihoods can be sustained in an urbanizing environment.

Hypothesis: A semi-urban land use network can complement rural livelihoods and reduce travel costs.

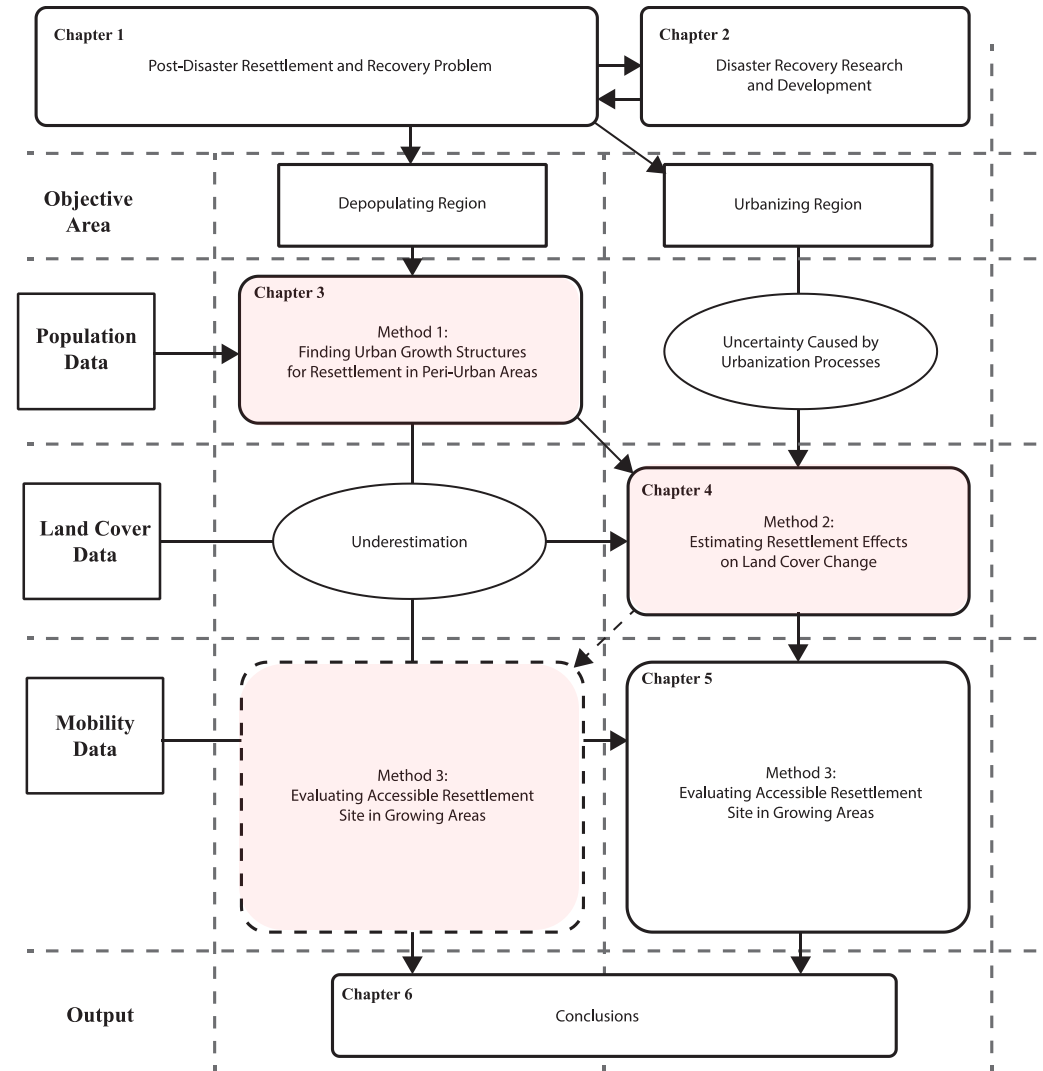
Results: Proved that higher transit increased access to livelihood options, thus reducing travel expenses.

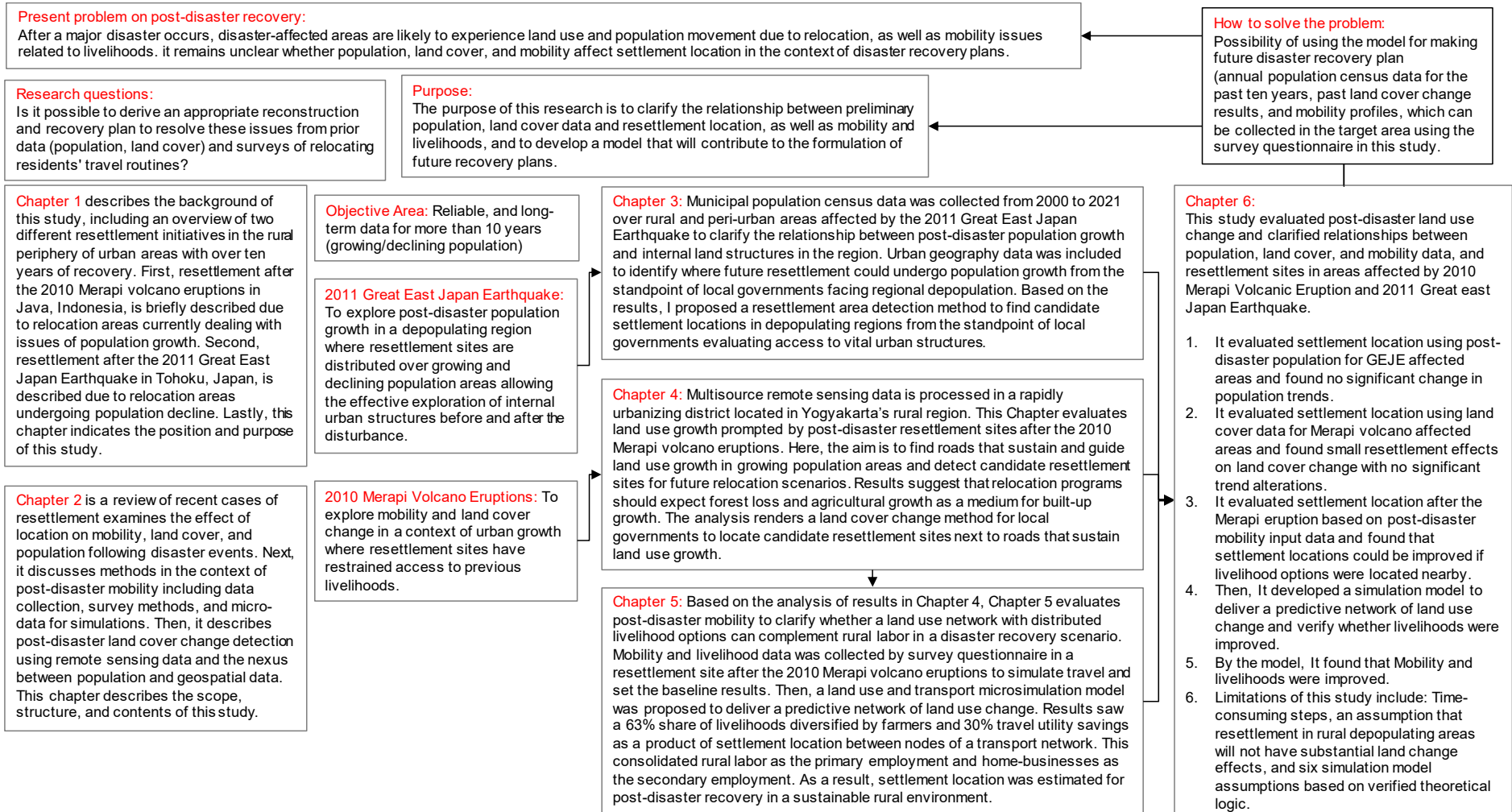
Findings: Rural livelihoods were sustained as the primary employment and home-businesses as the secondary employment.





- Future application of this model in a depopulating region is possible using **Chapter 3** and **Chapter 4's** result in **Chapter 5's** method as input data.
- Land use growth values from **Chapter 4** could be used as a safety measure to avoid land use dispersion if study areas are within the same ecozone.
- Admittedly, this study only referred to two cases. However, this research result was obtained by monitoring 10-years of recovery and there has never been any research like this before.
- I expect to investigate other cases in the future to improve the model evolving it into a tool that local governments will be able to use.





Chapter 2. Post-Disaster Recovery Research and Development.

Chapter 2 reviews a few cases of resettlement in the past 20 years that have been reported to lack mobility and livelihood facilities, land cover treatment, and have caused rural migration to peri-urban areas. It also describes the scope of this study underscoring gaps addressed.

Table 1. Post-Disaster Resettlement and Recovery Problems

Case	Relevance	Problem	Reference
2018 Sulawesi Earthquake	Resettlement forced part-time farmers to choose fishing over farming because of the many kilometers required to reach their farmlands (p.4).	Mobility	Paripurno et al., 2022
2011 Great East Japan Earthquake	Collective relocation projects resulted in the creation of a decentralized regional structure based on the separation of jobs and residences (p.16, Section 3.4, last paragraph).	Mobility	Tanaka, 2023
2010 Merapi volcano eruptions	Several households have to travel back to their farmlands daily or change their identities as farmers (p.202).	Mobility	Miller, 2018
2009 L'Aquila Earthquake	Resettled communities were isolated and made dependent on private transport to reach valued destinations and amenities (p. 80, Section 3, paragraph 1 & 3).	Mobility	Contreras et al., 2017
2009 L'Aquila Earthquake	Resettled households demonstrated a positive correlation between dissatisfaction and the distance and travel time to a place with employment.	Mobility	Contreras et al., 2013
2004 Indian Ocean Tsunami	Neuhuen relocation sites underscore the importance of infrastructure and mobility tied to livelihood recovery (p.180).	Mobility	Sina et al., 2019
2004 Indian Ocean Tsunami	Most people working in the capital city, Banda Aceh, spent more than 30% of their income on travel. Some jobs were location-reliant, such as farming and fishing industries, which meant that people had to commute to their previous villages for work (p.181).	Mobility	Sina et al., 2019
2004 Indian Ocean Tsunami	In Arugama bay, the eastern coast of Sri Lanka, low-income fishermen 2 kilometers away from the ocean could not foresee any livelihood options and had to walk 2 kilometers daily to sustain their families (p.609 Section 4.2).	Mobility	Ingram et al., 2006
2010 Merapi volcano eruptions	Settlements are not physically embedded in the nearby provincial capital city and challenge the official administrative boundary of cities promoting the expansion of built-up areas (p.200)	Land Cover	Miller, 2018
2004 Indian Ocean Tsunami	Resettlement camps have threatened the environmental conservation of Aruguma Bay, Sri Lanka (p.610).	Land Cover	Ingram et al., 2006
Bangladesh Floods 1995-2015	Land use change has exacerbated vulnerability in disaster risk areas and disasters have forced people's movement to periurban areas causing unprecedented vulnerable conditions and land cover degradation.	Land Cover	Parven et al., 2022
2011 Great East Japan Earthquake	Coastal-damaged areas changed from agricultural activities to grasslands. One third of risk zones were rendered unusable (Tanaka, 2023).	Land Cover	Ishihara and Tadono, 2017
2005 Hurricane Katrina	Built-up land disappeared and several structures were abandoned during the recovery phase. Land cover took over several roads and persistently revegetated the affected areas.	Land Cover	Reif et al., 2011
2011 Great East Japan Earthquake	Collective relocation projects resulted in the creation of low-density areas (p.16, Section 3.4, last paragraph).	Population	Tanaka, 2023
2011 Christchurch Earthquake	Massive outflow of people to the outer areas of the city, except people over 50 years of age representing a demographic group that tends to stay due to age and land ownership as well as a deeper connection to place (King and Gurtner, 2021, section 6.4.1).	Population	King and Gurtner, 2021
2005 Hurricane Katrina	Only 60% of the population returned to the city where most of the damage occurred (section 6.4.2). The other 40% were thinking of leaving before the event, but damage prompted their migration.	Population	King and Gurtner, 2021
Cydones Larry and Yasi, 2006 & 2011	The disaster exacerbated an already declining population trend due to the loss of land and property displacing most of the agricultural labor force (section 6.4.3).	Population	King and Gurtner, 2021

2-1-1 Mobility and Livelihoods

Disaster recovery research focuses on lives and livelihoods including mental health and redevelopment in socioeconomic terms (Contreras et al., 2016).

- For example, resettlement after the 2004 Indian Ocean tsunami underscored the importance of livelihood recovery tied to infrastructure and mobility options.
- Interviewed households in Neuheun relocations sites, Aceh Besar, Indonesia, stated that: *“infrastructure facilitates their mobility and connection to the workplaces and is essential for the socio-economic improvements of the relocated sites”* (Sina et al., 2019, p. 180).
- In another context, the 2009 L’Aquila earthquake in Italy resulted in the relocation of 19 communities to isolated settlements making people dependent on private transport to access jobs and amenities (Contreras et al., 2013, 2017).
- Figure 5 shows the core city of L’Aquila and the 18 resettlement sites detached from the city.

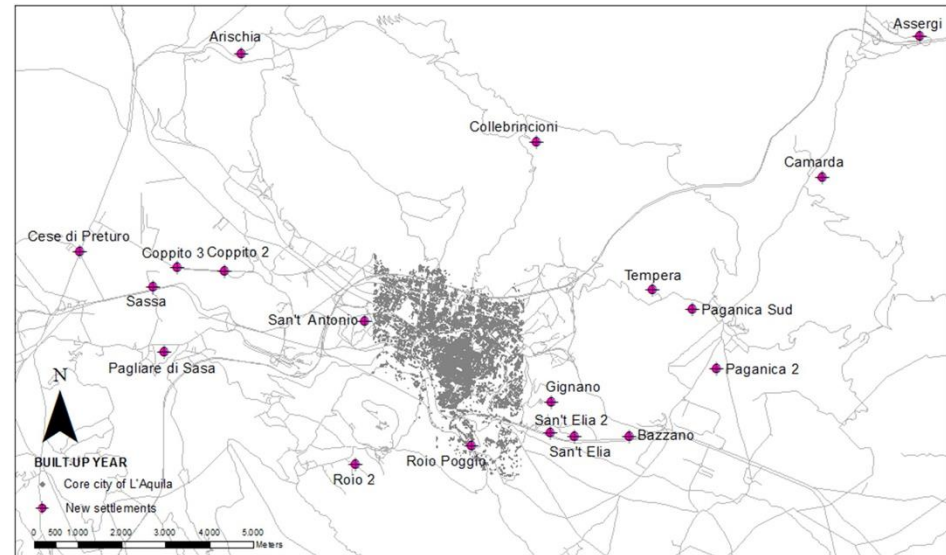


Figure 5. Detached resettlement sites after the 2009 L’Aquila Earthquake
Source: (No changes made, under license CC by 4.0 DEED, Contreras et al., 2017.
<https://doi.org/10.1016/j.techfore.2016.12.010>)

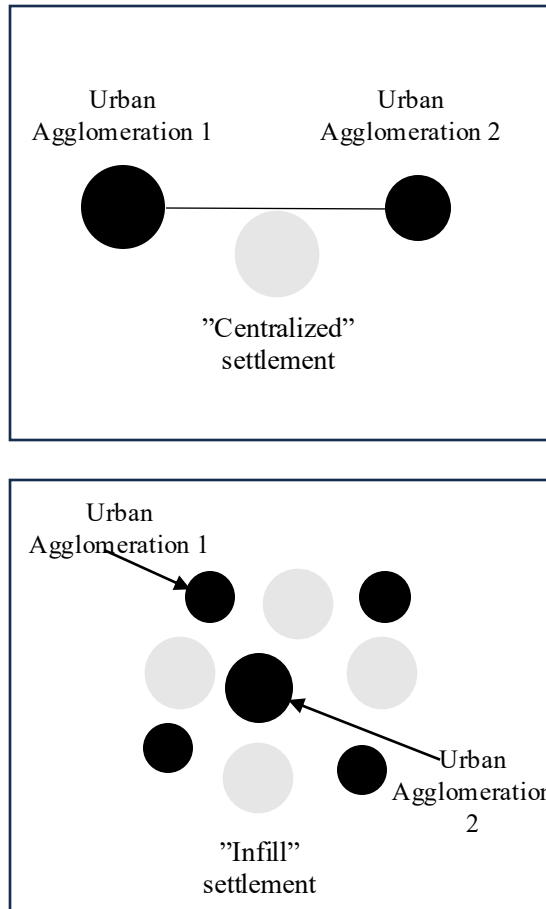


Figure 6. Spatial integration of centralized and infill resettlement types.

Source: (Adapted from [Hu and Ubaura, 2021](#))

2-1-2 Land Cover

In terms of land cover, successful relocation projects often identify equivalent lost land near new sites or bring new land into production ([Cernea, 2021](#)).

One example is the case of land scarcity near Shuikou dam in China, which led project leaders to convert hillsides and uplands into agrarian terraces ([Cernea, 2021](#)).

Another implication is the type of resettlement. Scattered settlements can waste land, generate poor living conditions, and degrade the environment ([Yi et al., 2012](#)).

A study by Hu and Ubaura ([2021](#)) found that “infill” and “centralized” resettlement sites in a host urban area were ranked best for spatial integration in China (Figure 6).

Centralized resettlement is considered an effective way to utilize rural land and improve infrastructure, public services, and living conditions ([Yi et al., 2012](#)).

2-1-3 Population

- Population growth in developing countries, where most urbanization is presently occurring, is equally contrasted by slow growth and population decline in developed countries ([King and Gurtner, 2021](#)).
- A disaster will prompt population movement in depopulating areas, but isolated resettlement sites can aggravate population loss if residential areas are separated from jobs, commerce, and amenities ([Tanaka, 2023](#); [Miyasada and Maly, 2021](#); [Abe, 2014](#)).
- For example, after hurricane Katrina in 2005, only 60% of the population returned to the affected areas ([King and Gurtner, 2021](#)).

Therefore, it remains unclear whether population, land cover, and mobility affect settlement location in the context of post-disaster recovery plans.

Novelty 1

- Population movement has gained the attention from researchers over the past years ([King and Gurtner, 2021](#)).
- Most studies in the post-disaster context report longitudinal fertility ([Lee et al., 2023](#)), mortality ([Miyazaki, 2022](#); [Nakasu, 2021](#); [Frankenberg et al., 2020](#)), and migration ([Abe, 2014](#)), but to the best of my knowledge, past studies have not explored the relationship between population growth and urban geography to clarify whether pre-disaster data can be used to detect future resettlement areas.

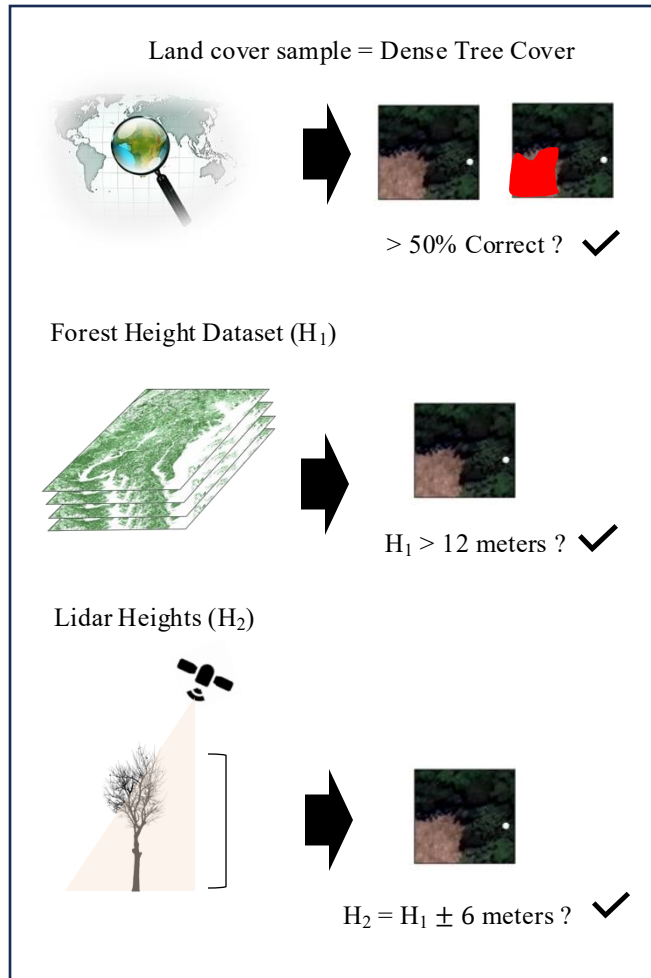


Figure 7. Three Quality Control Measures

- Monitoring land cover change from remote sensing data is essential to estimate resettlement and recovery investment effects.
- However, the predictive uncertainty of algorithms can stream inconsistent time-series data (Zhao et al., 2019).
- This problem is attributed to machine and deep learning models suffering from insufficient and inefficient labels (Huang et al., 2015):
 1. Supervising land cover classes is a time-consuming process and domain expertise is limited to particular sites (Li et al., 2022).
 2. Also, the reliability of labeled data can be contested due to interpretation biases (Foody, 2010; Stehman, 1999).
 3. Satellite images have multi-pixel geometric elements or mixed classes per pixel causing higher uncertainty (Xu et al., 2022).

One way to deal with these problems is using pre-existing land cover samples (Hermosilla et al., 2022). However, quality control measures are required.

Novelty 2

Figure 7 shows a unique method to automatically supervise pre-existing samples:

1. Three quality control measures are applied.
2. Reliable samples deliver accurate land cover class amplitude metrics to automatically supervise spectro-vertical features of candidate training samples.

- This study also estimates travel demand focusing on individual lifestyles, activity-based scheduling, and rescheduling (Bowman and Ben-Akiva, 2001).
- Figure 8 shows the framework of decisions to model travel demand (Ben-Akiva, Bowman, and Gopinath, 1996).
- However, human behavior is uncertain (Hagerstrand, 1970).
- Therefore, a new method was developed to circumvent this issue: Monte Carlo microsimulation (Moeckel et al., 2003).

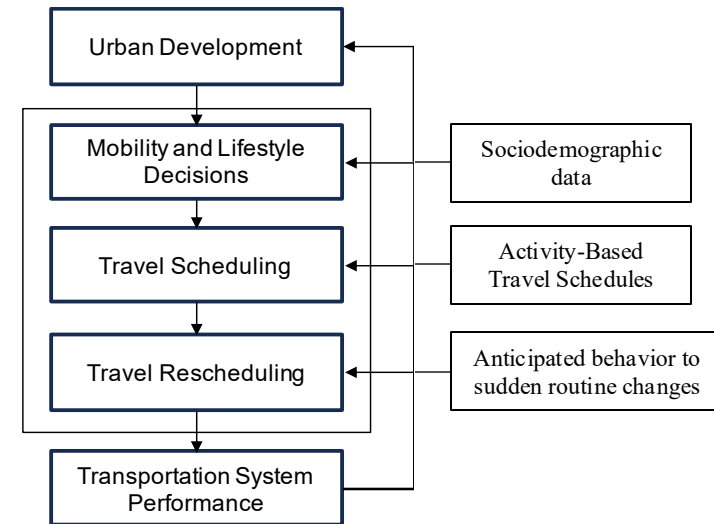


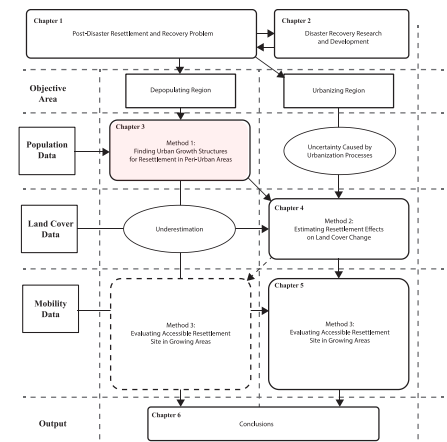
Figure 8. Disaggregate urban travel demand modeling system.
Source: (Adapted from Bowman and Ben-Akiva, (2001)

Novelty 3

Many influential studies have advanced the computational barriers inherent to microsimulation models (Miller and Salvini, 2005; Waddell et al., 2003; Wagner and Wegener, 2007; Zhu et al., 2018; Zondag et al., 2015), but they mainly derive findings from urban areas and do not address post-disaster mobility in rural areas.

Chapter 3. Evaluating Urban Growth Structures for Resettlement Area Detection after the 2011 Great East Japan Earthquake

This chapter evaluates the relationship between post-disaster population growth and land structures in Great East Japan Earthquake affected areas to identify where future resettlement could undergo population growth.



- On March 11, 2011, a 9.0 magnitude earthquake on the Richter scale caused a huge tsunami displacing >470,000 people from their homes.
- The percentage of population living inside the tsunami inundation area at the time of the event (Miyazawa, 2011) is shown in Figure 9 using 2010 census data.
- Population counts were extracted from the statistical census bureau of Miyagi prefecture [Dataset 1], Iwate prefecture [Dataset 2], Sendai city statistics [Dataset 3], and e-Stat (a government-led data website; [Dataset 4]).

20 municipalities were selected using two criteria:

1. Balance between highly-affected rias (11) and coastal plain municipalities (9).
2. Data availability for the analysis period (2000-2021).

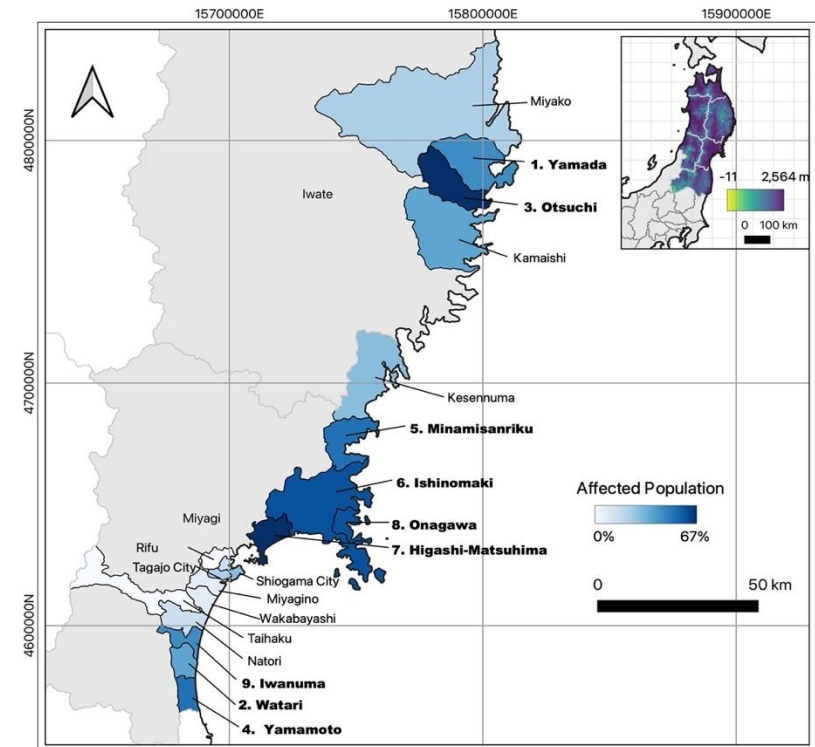


Figure 9. Percentage of affected population.

Data source: (Miyazawa, 2011. <http://tohokugeo.jp/articles/e-contents16.html>)

Population counts were plotted in Figure 10 to compare changes from 2000-2021.

Here, we can identify three trends:

- (1) “Severely affected” areas were declining before 2011.
- (2) Population movement as shown by aggravated population loss in 2011.
- (3) Sendai’s peri-urban area has stable/growing and declining municipal populations since before 2011.

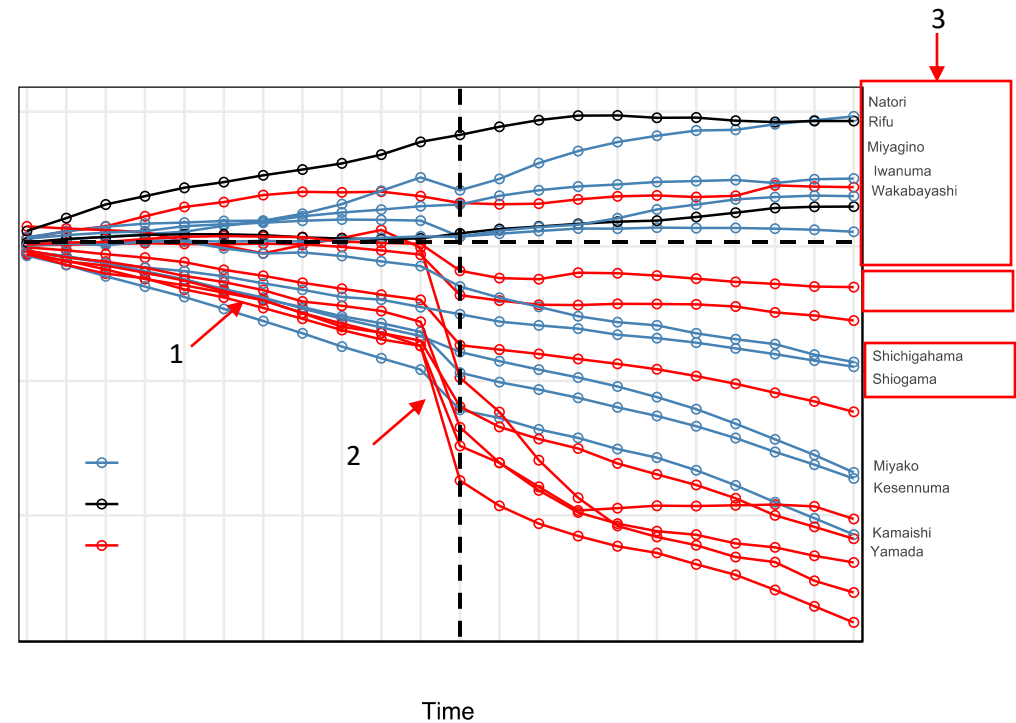


Figure 10. 2000-2021 Population trends. Values are in cumulative percentage of change.

Municipal population data was subdivided into pre- and post-disaster data groups, as well as growing and declining population trends.

Figure 11:

- (1) Exacerbated fertility loss after 2011.
- (2) Group 4 shows a steeper fertility loss after 2011.

Figure 12:

- (1) Groups 1-2 show stable mortality growth rates.
- (2) Group 4 shows abrupt mortality trend change.

Data Analysis:

- Exacerbated pre-disaster fertility rates, with no change in trend.
- Mortality in rias coast has seen a change in trend 5-6 years after the event.
- Mortality and in-migration were exceptionally large during the first 5 years of recovery.

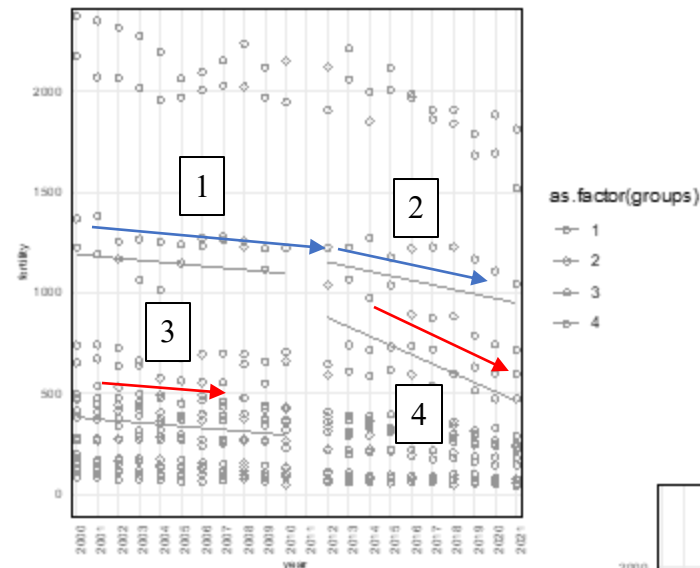


Figure 11. Fertility

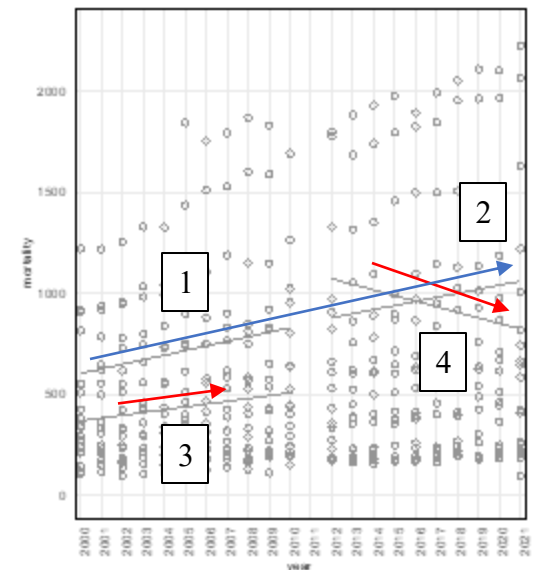
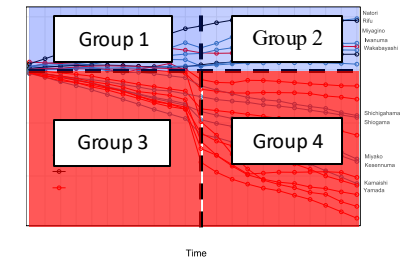


Figure 12. Mortality

- Urban geography features in Figure 13 were included to find correlations with population growth areas if any.
- A digital elevation model, MERIT-DEM ([Yamazaki et al., 2017](#)), was used to extract slope and elevation data, while the features below were extracted using a Geographic Information System.

Feature List

- Area in km² (GIS)
- Density (population / km²)
- Geography (Rias / Flat)
- Inundation rate in percentage ([IRIDeS, 2014](#))
- Ocean Front in linear meters (GIS)
- Euclidean distance from centroids to the closest main train station (GIS)

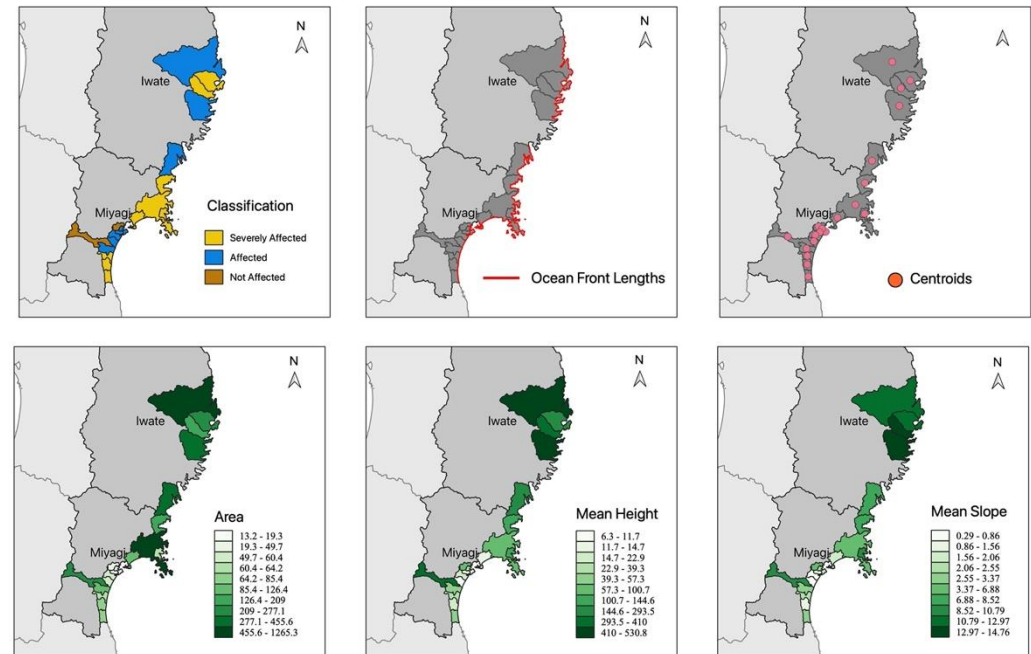


Figure 13. Geometric Features

3-5 Peri-Urban and Rural Population Comparison

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<i>Group 1: Peri-Urban</i>	<i>Coefficients</i>	<i>Standard Error</i>	<i>Min</i>	<i>Max</i>
Population Counts	-75145***	2423	-79994	-70296
Area	787***	81	625	950
Density	59***	2	55	63
Geography	2963	6917	-10878	16805
Ocean Front Length	1276***	246	784	1768
Mean Slope	5837**	2097	1641	10034
Dist. Train Station	148	365	-582	878

$R^2 = 0.99$

Observations = 66

Significant F-value

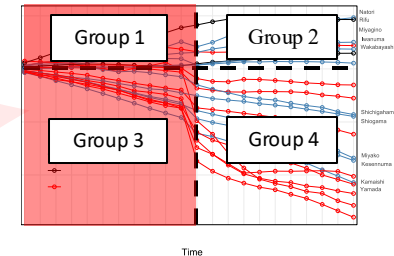
<i>Group 3: Rural</i>	<i>Coefficients</i>	<i>Standard Error</i>	<i>Min</i>	<i>Max</i>
Population Counts	11523***	2476	6629	16417
Area	22***	4	14	30
Density	12***	1	10	14
Geography	580	3110	-5567	6726
Ocean Front	572***	27	519	624
Mean Slope	-3032***	367	-3758	-2306
Dist. Train Station	2668***	428	1823	3514

$R^2 = 0.89$

Observations = 153

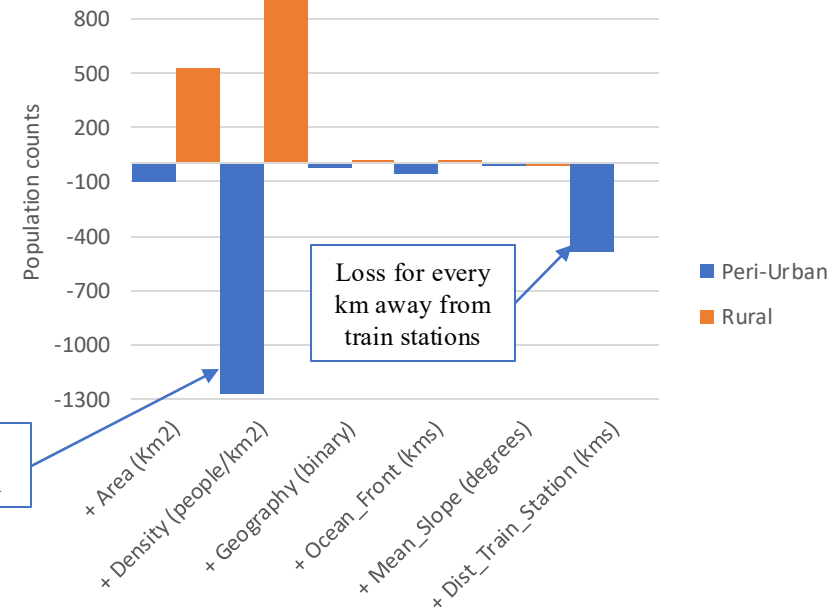
Significant F-Value

Peri-urban areas have sparsely distributed populations and rural areas have concentrated populations in smaller areas.



More people for every unit of density increase

Population per unit increase
(Before the 2011 Earthquake)



Less people for every unit of density growth

Loss for every km away from train stations

3-5 Peri-Urban and Rural Population Comparison

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<i>Group 2: Peri-Urban</i>	<i>Coefficients</i>	<i>Standard Error</i>	<i>Min</i>	<i>Max</i>
Population Counts	-74785***	7206	-89238	-60332
Area	720**	226	267	1173
Density	64***	5	53	74
Geography	-9516	18916	-47457	28425
Ocean_Front_Length	784	650	-519	2087
Mean_Slope	9421*	5591	-1793	20634
Dist_Train_Station	-937	861	-2665	790

$R^2 = 0.96$

Observations = 60

Significant F-Value

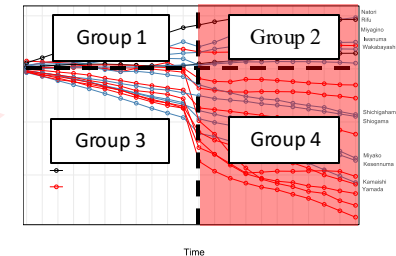
<i>Group 4: Rural</i>	<i>Coefficients</i>	<i>Standard Error</i>	<i>Min</i>	<i>Max</i>
Population Counts	7445***	2149	3194	11696
Area	14***	4	7	21
Density	13***	1	11	15
Geography	-2184	2706	-7536	3167
Ocean_Front	637***	22	593	681
Mean_Slope	-3089***	319	-3720	-2458
Dist_Train_Station	2941***	374	2202	3680

$R^2 = 0.94$

Observations = 141

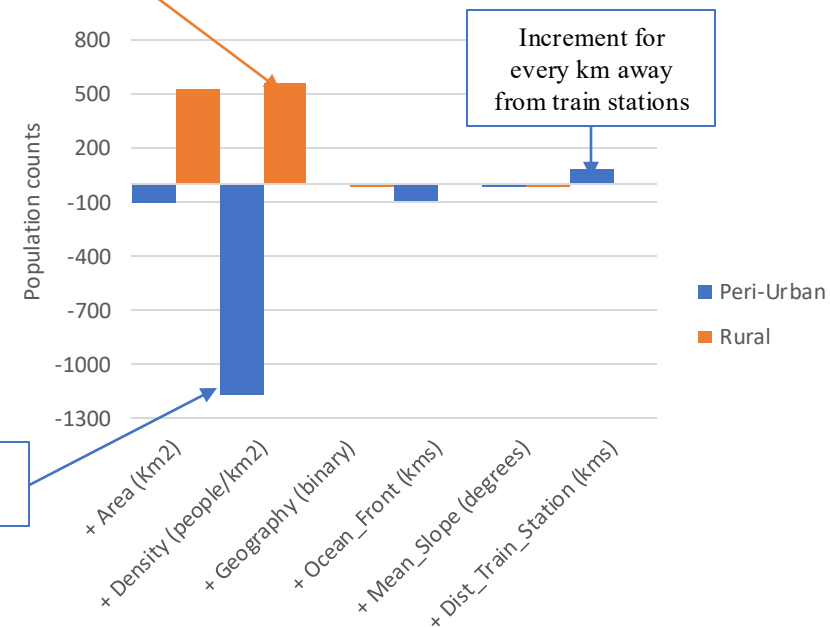
Significant F-Value

Peri-urban areas underwent small but dispersed growth. Rural areas saw significant loss of density.

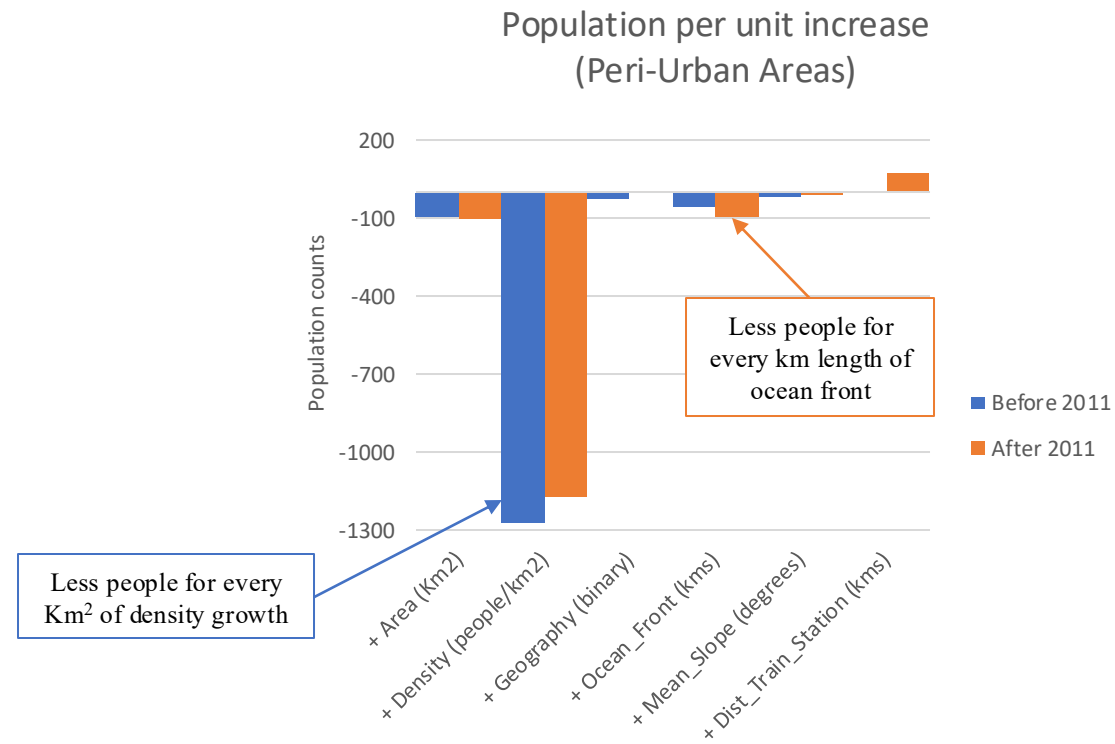
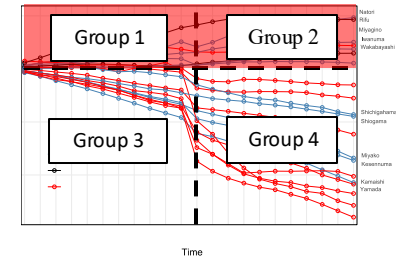


Significantly less people for every unit of density increase

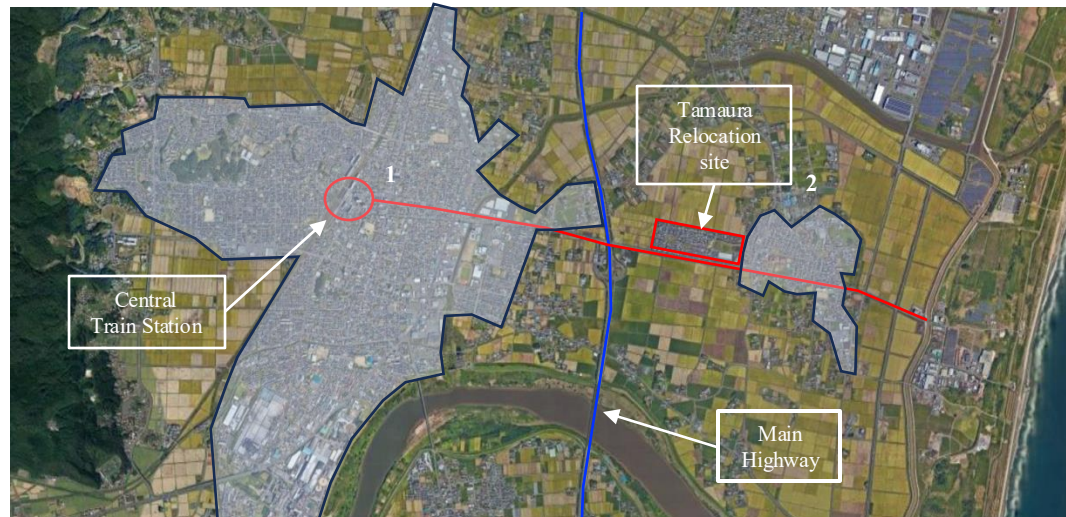
Population per unit increase (After the 2011 Earthquake)



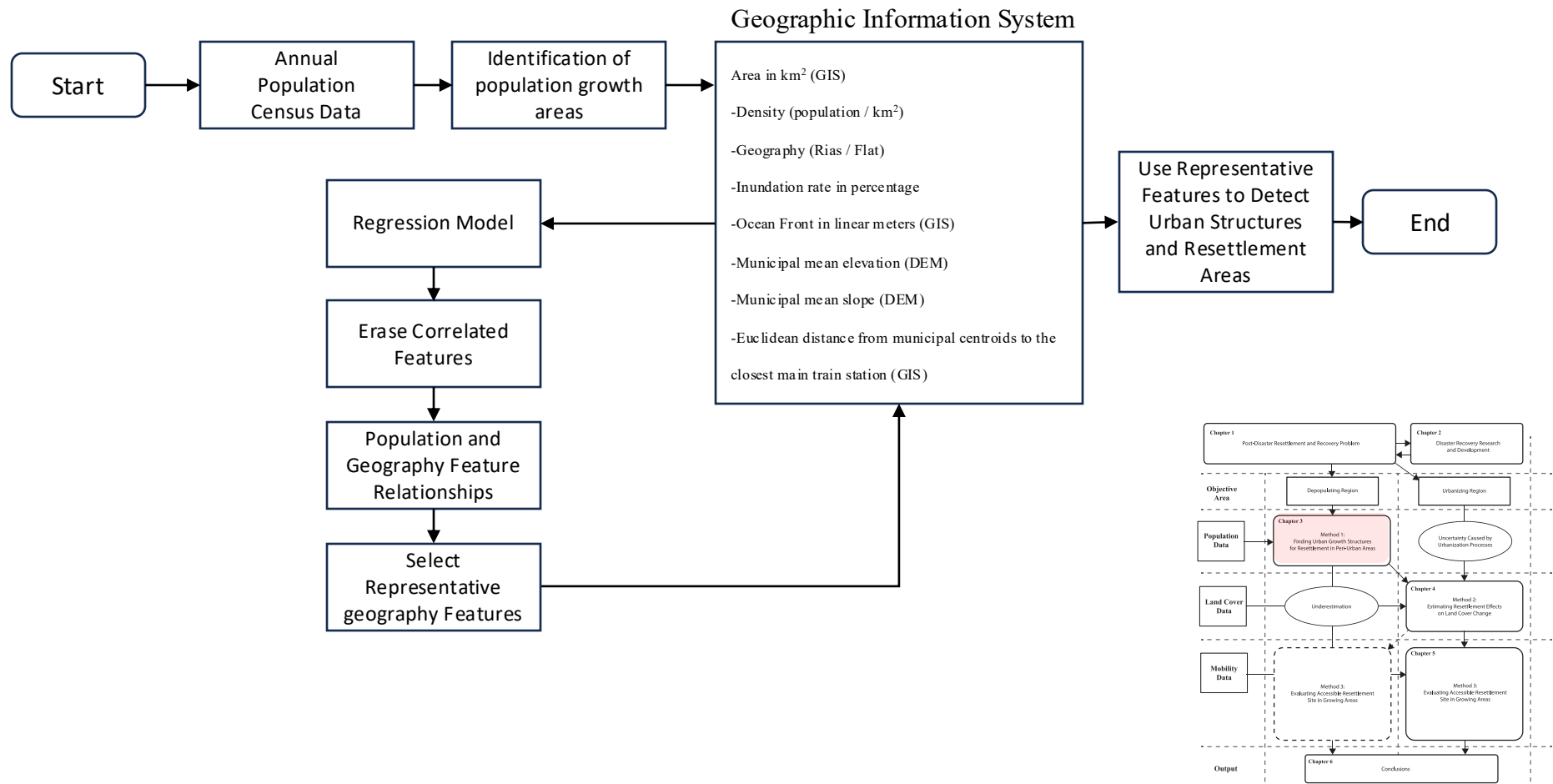
- I applied another regression to detect relationships between Group 1 and Group 2.
- Population growth in peri-urban areas was shown to increment with distance away from train stations.
- While this may seem irregular, it resonates with public transport in a decentralized regional structure, highlighted by Tanaka (2023) to be one of the major challenges for sustaining train operations.
- It also shows less people for every kilometer length of ocean front with less than 100 people in average.
- The sensibility of feature relationships were significant at 99% and all regressions had an R^2 of at least 89%.



1. Natori, Wakabayashi, and Iwanuma municipal areas are moderately growing.
2. I found that post-disaster population density increased in sparsely distributed peri-urban areas with a preference for vehicular accessibility.
3. 41% of the population in Iwanuma lived inside the tsunami inundation range making it more reliable for urban feature analysis than Natori (14%) and Wakabayashi (5%).
4. Therefore, Iwanuma was analyzed. It has two built-up agglomerations and peripheral farmlands. Tamaura relocation site is located in-between the two built-up agglomerations and is proximal to the main highway.
5. Findings suggest that overall pre-disaster trends were unchanged. Also, I found that vehicular accessibility and built-up density surrounded by farmlands play an important role in the stability of population growth areas. This is not a common measure of urban growth and can be used to detect future resettlement areas.

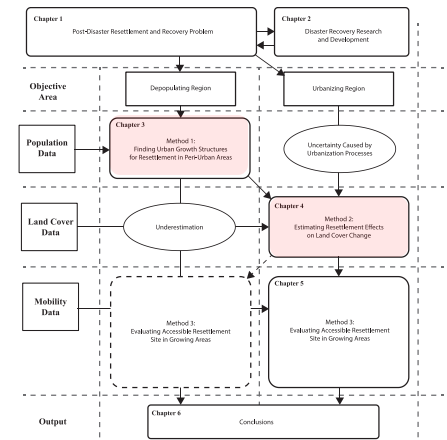


A resettlement area detection method was rendered to find peri-urban areas likely to experience urban growth in depopulating regions.



Chapter 4. Automatic Supervision of Land Cover Labels to Increment the Reliability of Training Datasets for Disaster Recovery Change Detection

Chapter 4 evaluates land change interactions prompted by post-disaster resettlement sites in a context of urban growth to help local governments identify relocation sites next to roads that sustain land use growth.



4-1 The 2010 Merapi volcano eruptions, Yogyakarta, Indonesia

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- On October, 2010, the Merapi volcano erupted thrice displacing thousands people from their homes ([Kuncoro et al., 2012](#)).
- The government resettled 75% of displaced people in 12 resettlement sites sparsely distributed over the Cangkringan district (Figure 14).
- Figure 14 shows areas of influence drawn from the centroid of each settlement to represent the assumed daily median walking distance, equivalent to 800 m ([Yang and Diez-Roux, 2012](#)).

Then, 7 resettlement sites were selected using three criteria:

1. Heterogenous geographies.
2. Settlements in urban and rural areas.
3. No overlapping influence areas.

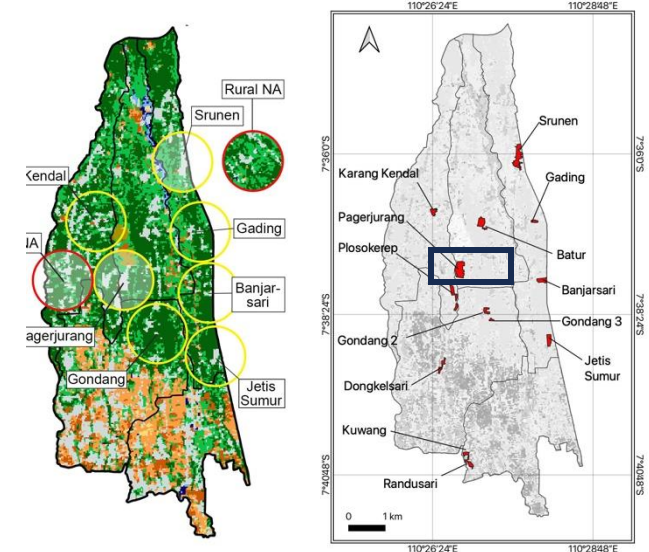
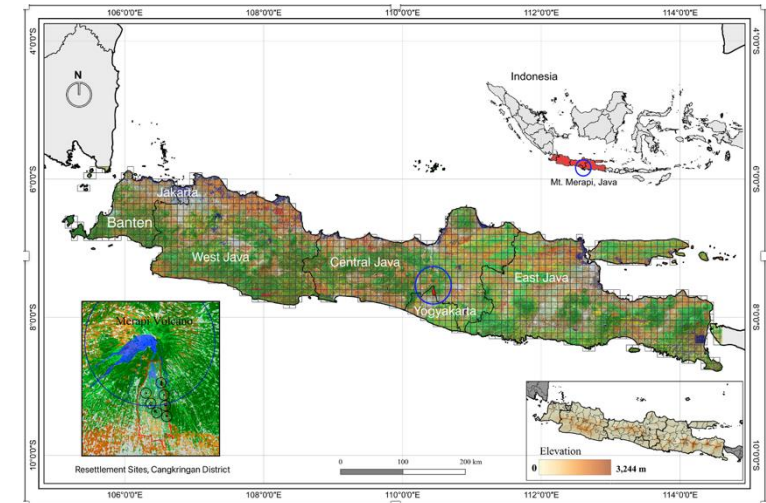


Figure 14. Resettlement sites in the Cangkringan district. The inset box shows the target area analyzed used in the next chapter.

1. ICESat-2 Lidar data

([Nueuenschwander and Pitts, 2019](#))

172,000 lidar shots were acquired to sample multisource remote sensing data in Table 4.

2. Global Land Cover and Land Use for 2019

([Hansen et al., 2022](#))

Was selected to improve upon its land use accuracy.

3. Multi-Error Removed Improved-Terrain DEM

([Yamazaki et al., 2017](#))

MERIT-DEM was selected based on its accuracy.

4. Multi-Spectral Sentinel-2 Satellite Imagery

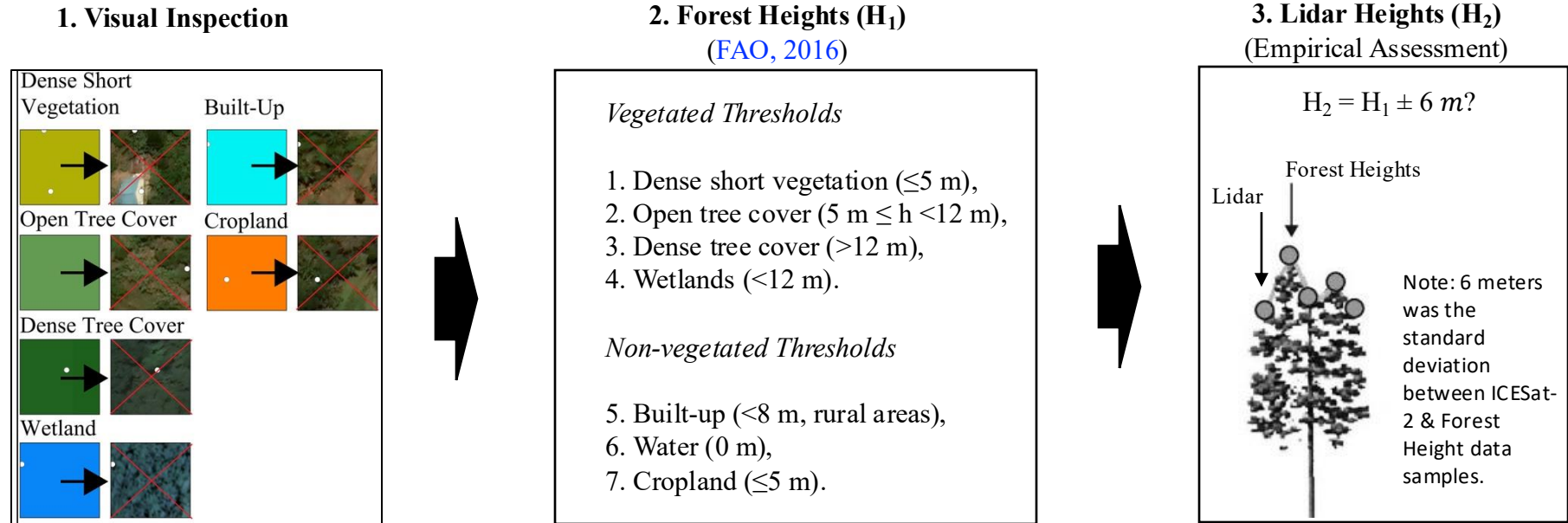
([ESA, 2015](#))

2016-2021 annual satellite images were collected from Google Earth Engine ([Gorelick et al., 2017](#)), pre-processed, and mosaicked into image composites west (~66 Million pixels) and east (~79 Million pixels).

Table 4. Data collected

Type	Resolution	Acronym	Dataset	Description/Formula	Reference
Land Cover	~30 m	GLCLU	Global Land Cover Land Use, 2019	Global land cover product	(Hansen et al., 2022)
Forest Heights	~30 m	Forest Heights	Global Forest Canopy Heights, 2019	GEDI-derived forest heights	(Potapov et al., 2021)
DEM	~90 m	MERIT-DEM	Multi-Error-Removed Improved-Terrain DEM	Digital Elevation Model	(Yamazaki et al., 2017)
	~90 m	Slope	DEM-Derived Slope	Ancillary Dataset	
	~90 m	TSRI	Transformed Solar Radiation Index	Ancillary Dataset	(Roberts and Cooper, 1989)
	~90 m	TWI	Geomorpho90m- Topographic Wetness Index	Ancillary Dataset	(Amatulli et al., 2020)
Sentinel-2 Spectral Indices	~10m, 20m	TCT	Tasseled Cap Transformation	Coefficients for TCG and TCW	(Shi and Xu, 2020)
	~10m, 20m	TCG	Tasseled Cap Greenness	$-0.3599*B - 0.3533*G - 0.4734*R + 0.6633*NIR - 0.0087*SWIR1 - 0.2856*SWIR2$	(Kauth and Thomas, 1976)
	~10m, 20m	TCW	Tasseled Cap Wetness	$0.2578*B + 0.2305*G + 0.0883*R + 0.1071*NIR - 0.7611*SWIR1 - 0.5308*SWIR2$	
	~10m	NDVI	Normalized Difference Vegetation Index	$NIR - R / NIR + R$	(Tucker, 1979)
	~10m	GNDVI	Green Normalized Difference Vegetation Index	$NIR - G / NIR + G$	(Gitelson et al., 1998)
	~10m	NDWI 1	Normalized Difference Water Index 1	$NIR - SWIR1 / NIR + SWIR1$	(Gao, 1996)
	~20m	NDWI 2	Normalized Difference Water Index 2	$G - NIR / G + NIR$	(McFeeters, 1996)
	~10m, 20m	RS2	Normalized Difference Flooding Index	$R - SWIR2 / R + SWIR2$	(Boschetti et al., 2014)
	~20m	NS2	NIR-SWIR2	$NIR - SWIR2 / NIR + SWIR2$	(Potapov et al., 2021)
	~10m	SAVI	Soil-Adjusted Vegetation Index	$(NIR - R / NIR + R + 0.5) * 1.5$	(Huete, 1988)
	~10m, 20m	IREFI	Inverted Red-Edge Chlorophyll Index	$(B7 - R) / (B5 / B6)$	(Frampton et al., 2013)
	~10m	BN	Reference index	$B - NIR / B + NIR$	(Potapov et al., 2020)
	~20m	S12	Reference index	$SWIR1 - SWIR2 / SWIR1 + SWIR2$	(Potapov et al., 2020)

First, pre-existing land cover samples were vetted against three quality control mechanisms to retain high-quality samples:



Each mechanism assigned one point to each correctly labeled sample. Then, the cumulative score of each sample rendered the level of reliability.

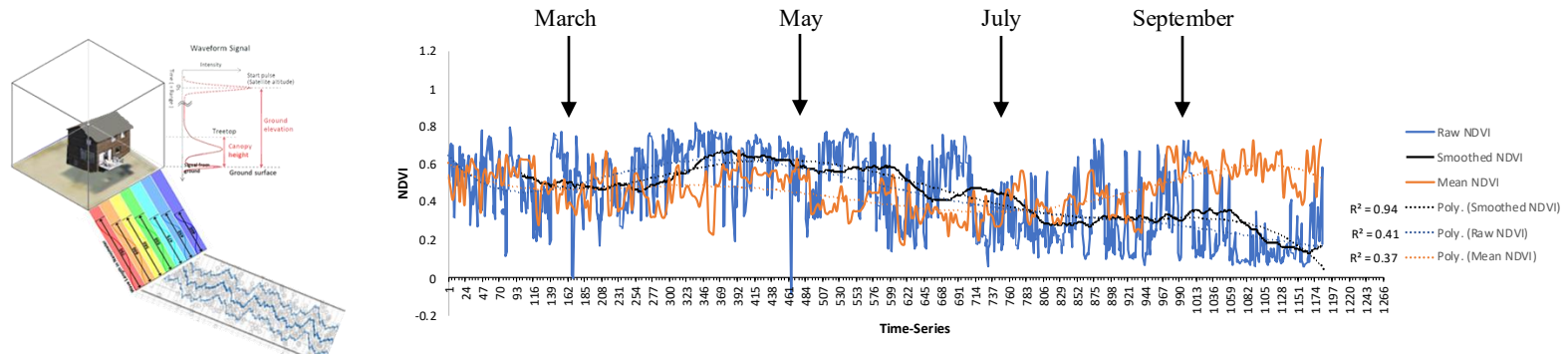
L3 = The highest quality samples

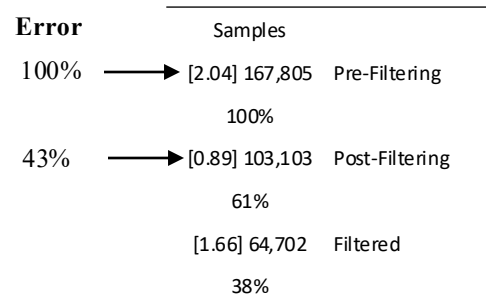
L2 = Medium quality samples

L1 = Low quality samples

L0 = Errors

- 500 m distributed L3 samples were held for accuracy assessments, and the prevailing L3 and L2 samples were compiled in a dataset.
- I used the dataset to sample multitemporal (satellite) data and topographical (DEM-derived) data.
- The resulting land cover class data sets were used to extract the statistical bounds (+/- standard error) of feature amplitudes (Figure 16).
- These metrics were used to detect the feature space of candidate training samples replacing the need for visual inspection in the region.





- A total of 167,805 candidate samples were automatically supervised rendering 103,103 reliable samples with 57% of the total error eliminated from the set.
- Refined samples were used to train and test 10 Random Forest models with accuracies shown in Table 5.

Table 5. Parameters and test data accuracy.

Pre-Selected Models								
	(mtry, depth, min.samp.split)	(#feat, max_feat.)	max samples	RMSE	Test Data Accuracy	Bias-squared	Variance	Total error
Model 1	(391, 17, 23)	(18, log2)	0.8	3.12	0.78 (0.005)	7.58	0.64	8.22
Model 2	(391, 19, 15)	(18, log2)	0.8	3.19	0.73 (0.005)	7.77	0.78	8.55
Model 3	(391, 19, 16)	(18, log2)	0.8	3.21	0.73 (0.005)	7.87	0.61	8.48
Model 4	(391, 19, 15)	(18, log2)	0.8	3.19	0.73 (0.004)	7.97	0.56	8.53
Model 5	(391, 17, 23)	(18, log2)	0.8	3.07	0.78 (0.003)	7.94	0.75	8.69
Model 6	(391, 17, 19)	(18, log2)	0.8	3.31	0.78 (0.004)	7.68	0.56	8.24
Model 7	(391, 17, 22)	(18, log2)	0.8	3.37	0.84 (0.002)	7.44	0.49	7.93
Model 8	(390, 17, 25)	(18, log2)	0.8	3.37	0.79 (0.003)	7.81	0.63	8.44
Model 9	(391, 17, 20)	(18, log2)	0.8	3.35	0.78 (0.003)	8.07	0.66	8.73
This Method	(391, 17, 22)	(18, log2)	0.8	3.08	0.78 (0.005)	8.11	0.61	8.72

Note: Model 7 presented higher accuracy and lower error but predicted biased results.

- I classified Java's east composite and performed accuracy assessments using two designs.
- Design 1: 269 labeled samples with visual inspection.
- Design 2: 269 L3 samples held for accuracy assessment.
- L3 samples using Design 2 exhibit 34% less error than samples using Design 1. Therefore, Design 2 was selected.
- Final-stage post-processing steps were subsequently performed increasing the overall accuracy from 78% to 82% as shown in Figure 17.

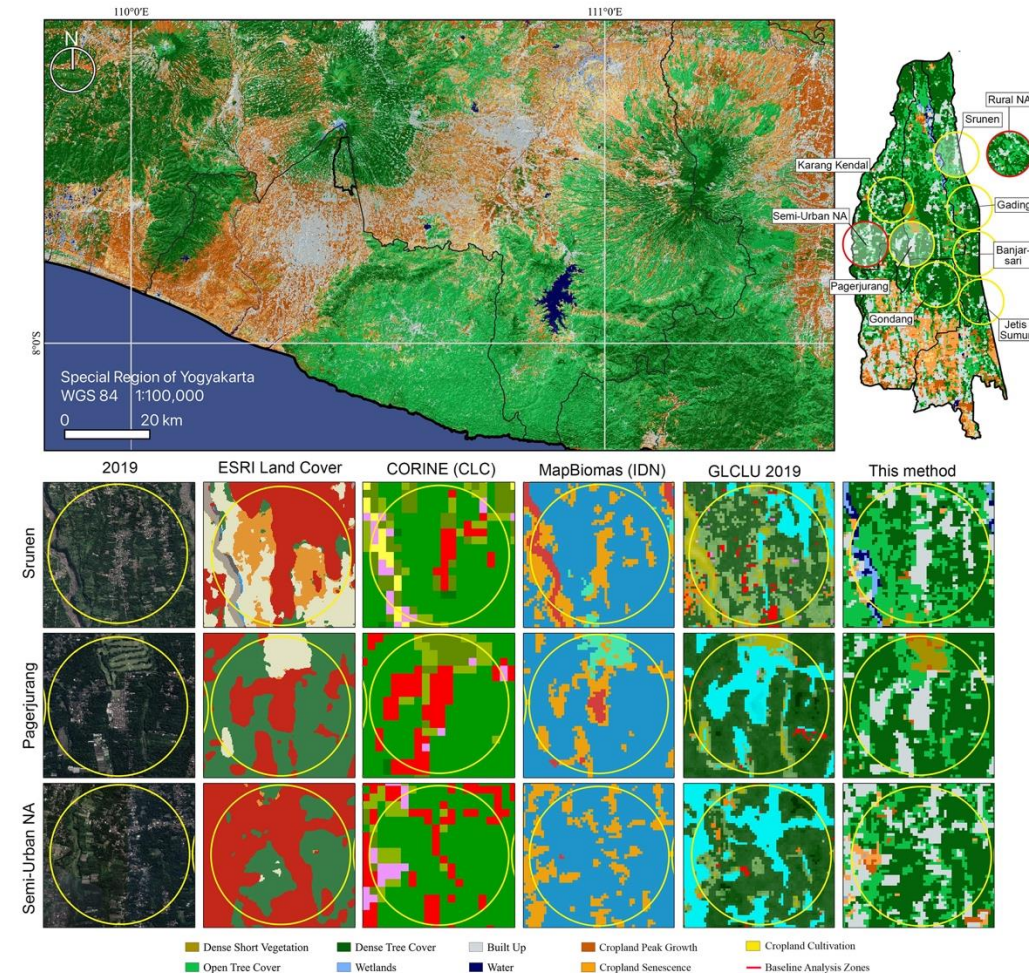


Figure 17. Method accuracy in comparison to available land cover products.

- Resettlement sites are targeted on this evaluation.
- **I estimated 2016-2021 annual land cover change data at the district and settlement level.**
- Table 6 shows forests having the highest net change values in sub-districts (4.5%) and resettlement areas (-6.9%).
- Conversely, built-up growth denotes 0.3% on the aggregate level and 4.7% in localized areas.
- In particular, cropland growth was seen in resettlement areas and cropland loss in sub-districts.
- Therefore, agriculture has grown near resettlement sites despite the overall declining tendency.

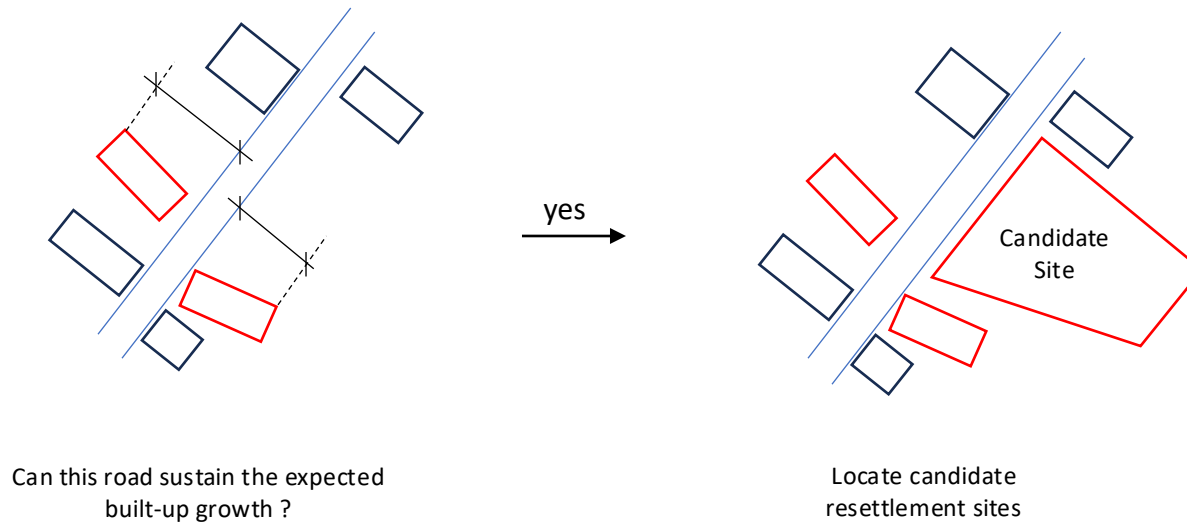
Table 6. Land change interactions.

		Crop Change	Built Change	Forest Change
Sub-District	Glagaharjo	-16.20%	3.00%	13.20%
Sub-District	Kepuharjo	3.10%	-9.10%	6.00%
Sub-District	Umbulharjo	-1.40%	7.10%	-5.80%
Settlement	Srunen	4.60%	8.30%	-12.90%
Settlement	Gading	2.00%	-2.50%	0.50%
Settlement	Banjarsari	2.50%	8.20%	-10.80%
Settlement	Jetis Sumur	10.30%	4.30%	-14.50%
Settlement	Pagerjuran	-2.30%	2.80%	-0.40%
Settlement	Karang Kendal	-1.80%	11.60%	-9.80%
Settlement	Gondang	0.30%	0.00%	-0.30%
Mean Total	Sub-Districts	-4.80%	0.30%	4.50%
	Settlements	2.20%	4.70%	-6.90%

Results suggest that future settlements in the region should expect forest loss and agricultural growth as a medium for built-up growth with:

16.4% forest loss,
18.5% cropland growth, and
15.5% built-up growth of the district's total land change.

These results provide land change estimates and sequences to help local governments find resettlement sites next to roads that sustain land use growth and avoid land cover loss.

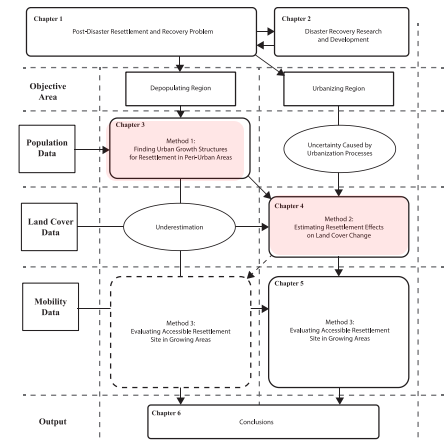
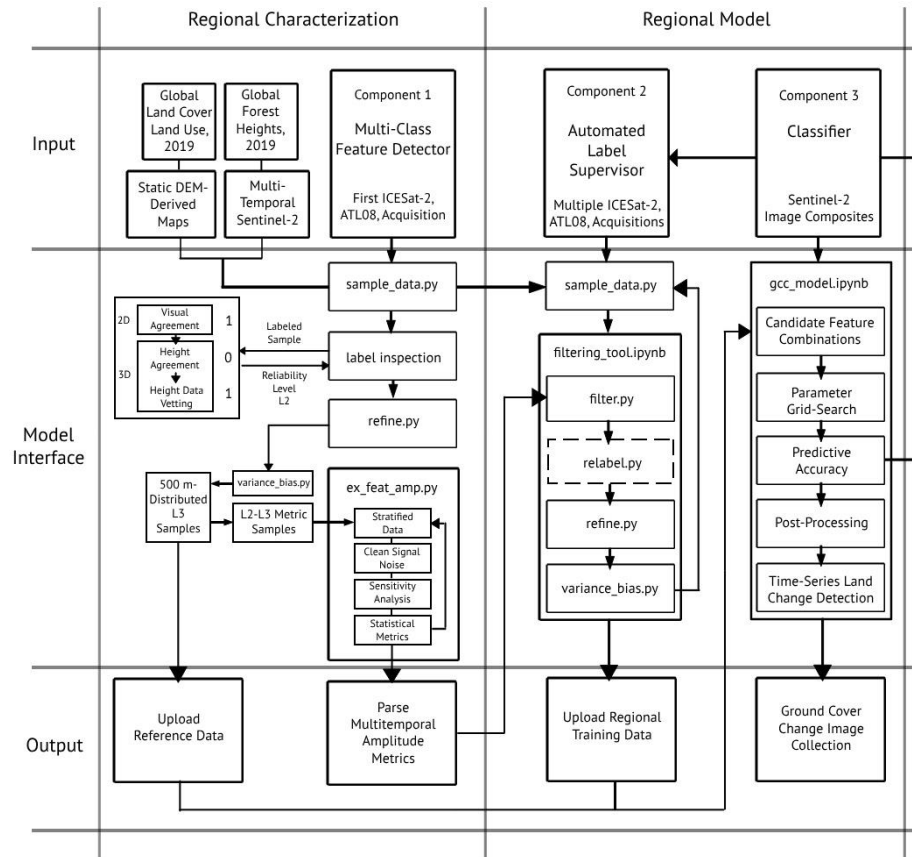


4-8 Method 2: Ground Cover Change (GCC) Model

42

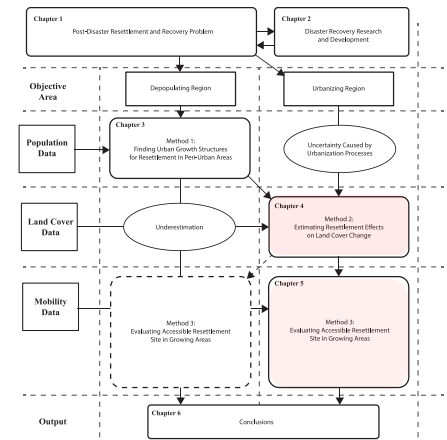
The proposed method can also:

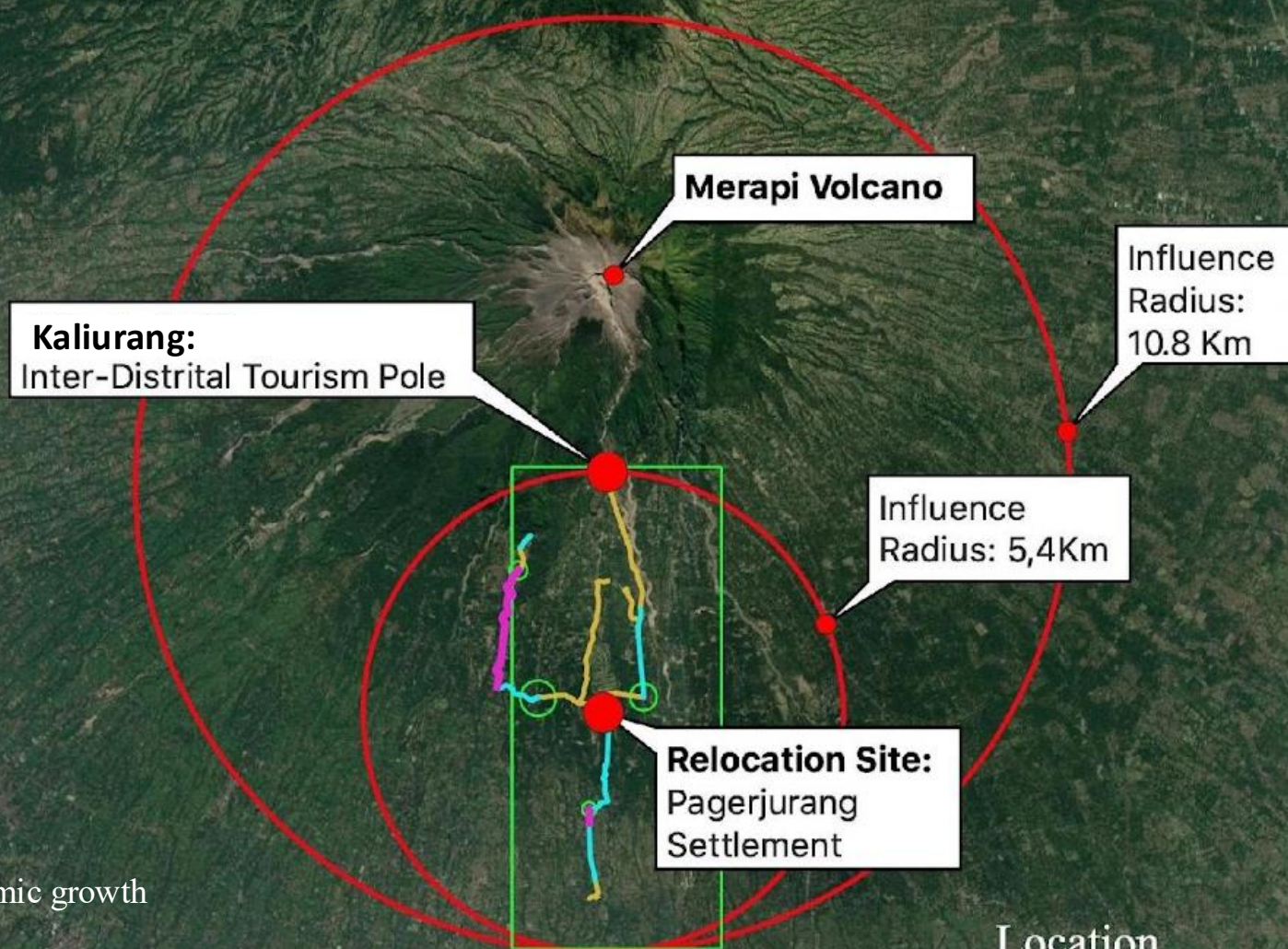
- Monitor socioeconomic effects of recovery through land change detection.
- Monitor investments to anticipate where readjustments need to be made.
- Evaluate ecological impacts and spatial needs.



Chapter 5. Land Use Microsimulation Model for Livelihood Diversification after the 2010 Merapi Volcano Eruptions

This chapter takes Chapter 4's result to find a candidate site. Then, it evaluates post-disaster mobility to clarify whether a land use network with distributed livelihood options can complement rural labor in a disaster recovery scenario.

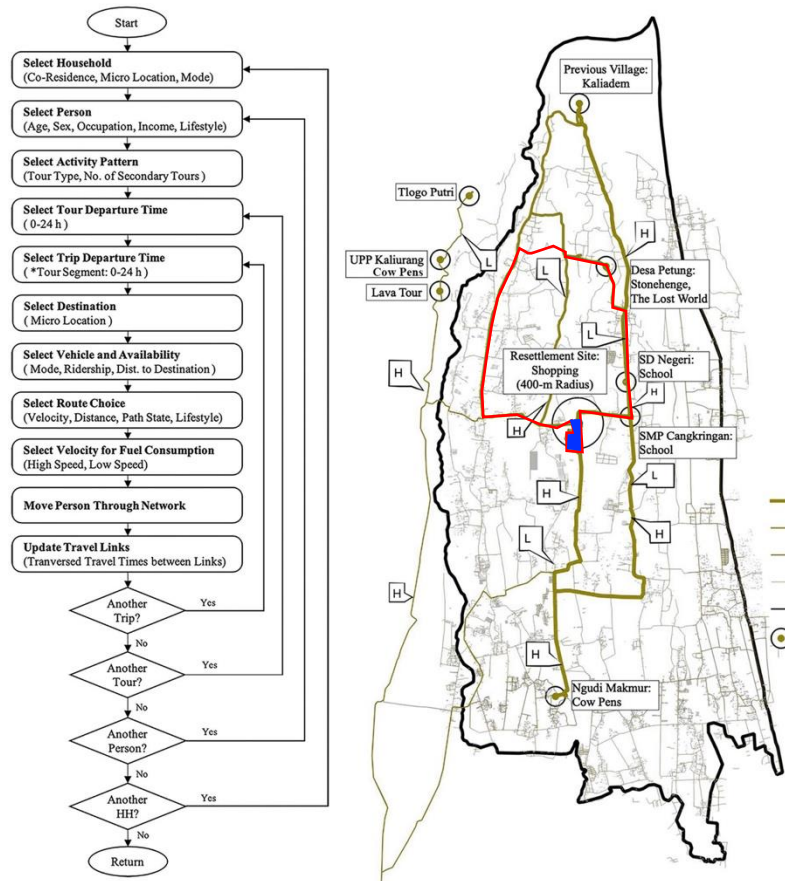




First, I identified economic growth areas.

Kaliurang, a small town, and Yogyakarta city representing two economic poles (Pramana, 2016).

Then, the local road network was surveyed to identify road hierarchies, road widths, and connectivity.



- Figure 19 shows a small road network found in red and a detected resettlement site next to the network in blue.
- I proceeded to collect mobility and livelihood data by survey questionnaire in the selected resettlement site.
- Travel schedules were used to locate travel destinations in GIS and start a travel simulation:
 1. Travel route choices were based on the shortest path with the least number of directional changes (Peponis et al., 2008).
 2. Sectoral speed identifiers, high (H) or low (L), were incorporated every 100–200 m, depending on roads (either paved or non-paved).
 3. The operation cost of a motorcycle was calculated with fuel price at 0.49 USD/Liter during the survey with monthly maintenance of 14.00 USD (Herwangi et al., 2017).

Figure 19. Road Network and Travel simulation.

- A total of 41 Javanese subjects were interviewed over three days, for 1–2 h each.
- The collected data included (see Table 7 for reference):
 1. A short census: Age, sex, and household size.
 2. Sociodemographic data: Marital status, hobby, job type, housing preferences for peri-urban areas (two types), and a lifestyle questionnaire for developing countries.
 3. Weekday travel routines: Time of departure, origin, exact destinations throughout the day, and travel modes (walking, car, motorbike, etc.).

Table 7. Sociodemographic data collected by field survey.

No.	Household (HH)	Adults per HH	Members	Job Type	Acronym	Segmentation	Acronym	Car	motorbike	Income/month (IDR)	Housing pref (1)	Housing pref (2)	Age
1	1	2	1	farmer	SI	Progressist	P	-	1	35	attractiveness	social neighborhood	38
2			2	livestock	SI	modern	M	-	0	105	attractiveness	social neighborhood	40
3	2	1	1	livestock	SI	modern	M	-	1	-	-	social neighborhood	65
4	3	1	1	merchandising sales	C	Progressist	P	-	1	105	-	social neighborhood	41

1. The first travel simulation evaluated settlement location after the Merapi eruptions based on post-disaster mobility input data.
2. Results revealed a restrained potential to increase household profits after utilizing 26% of the average monthly income for travel.
3. This was exacerbated by the limited access to livelihood options.
4. However, settlement location could be improved with livelihood options nearby.
5. So, I proposed a land use microsimulation method.

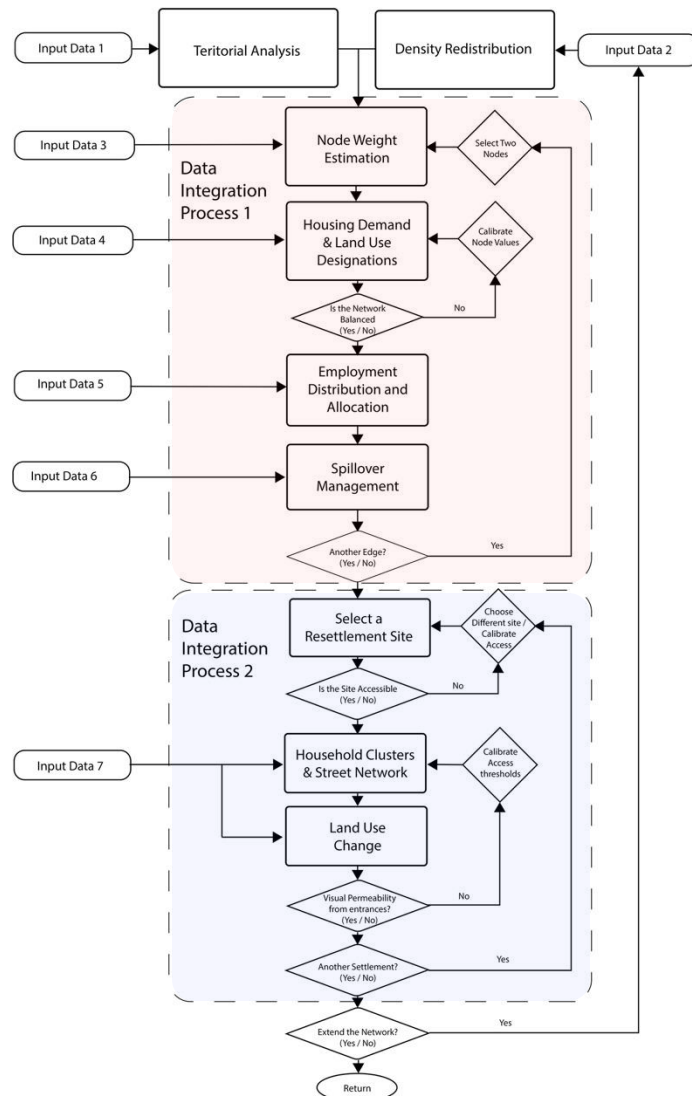


Figure 20. Method Components

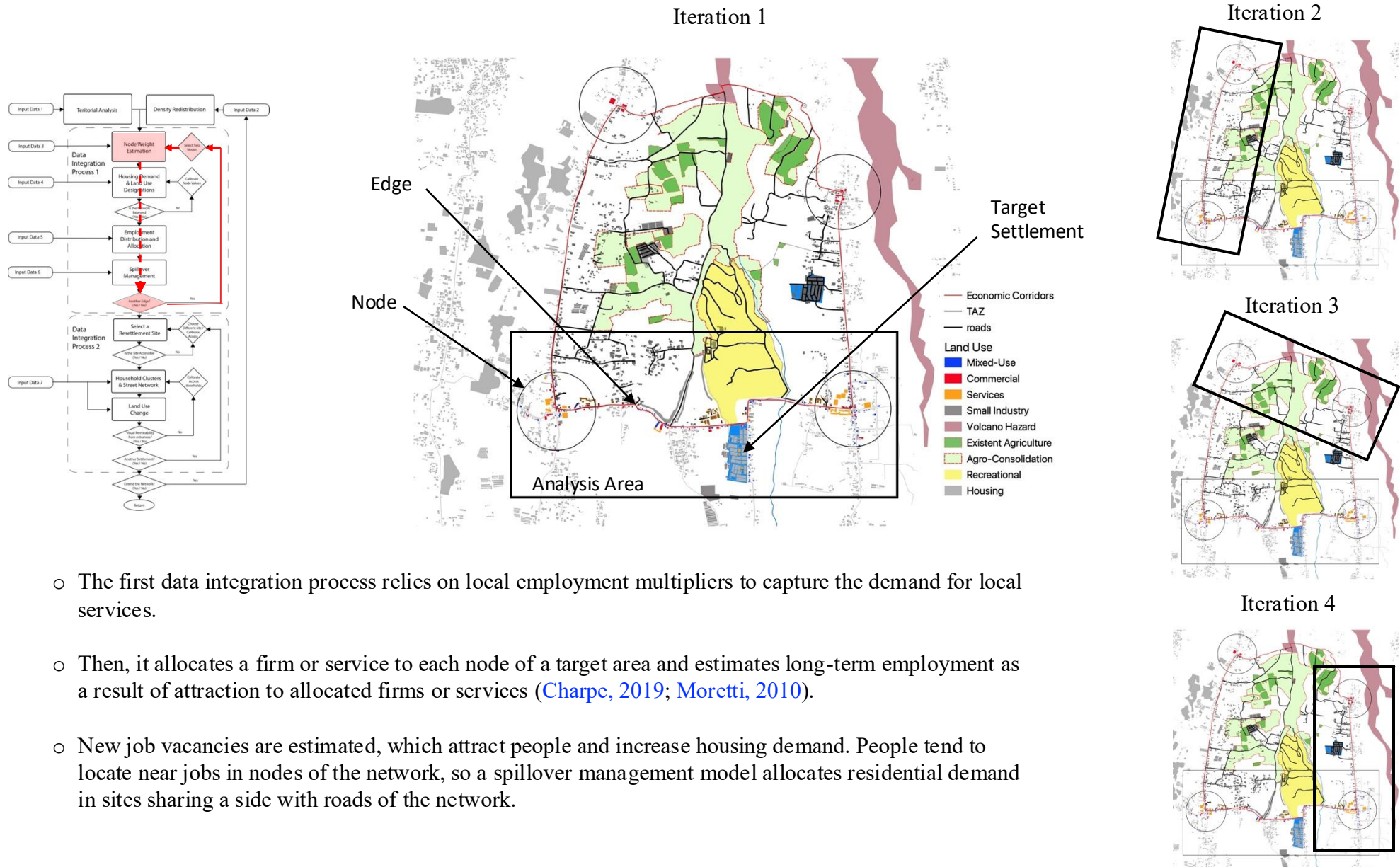
Here, I briefly describe method 3 shown in Figure 20.

Data integration process 1:

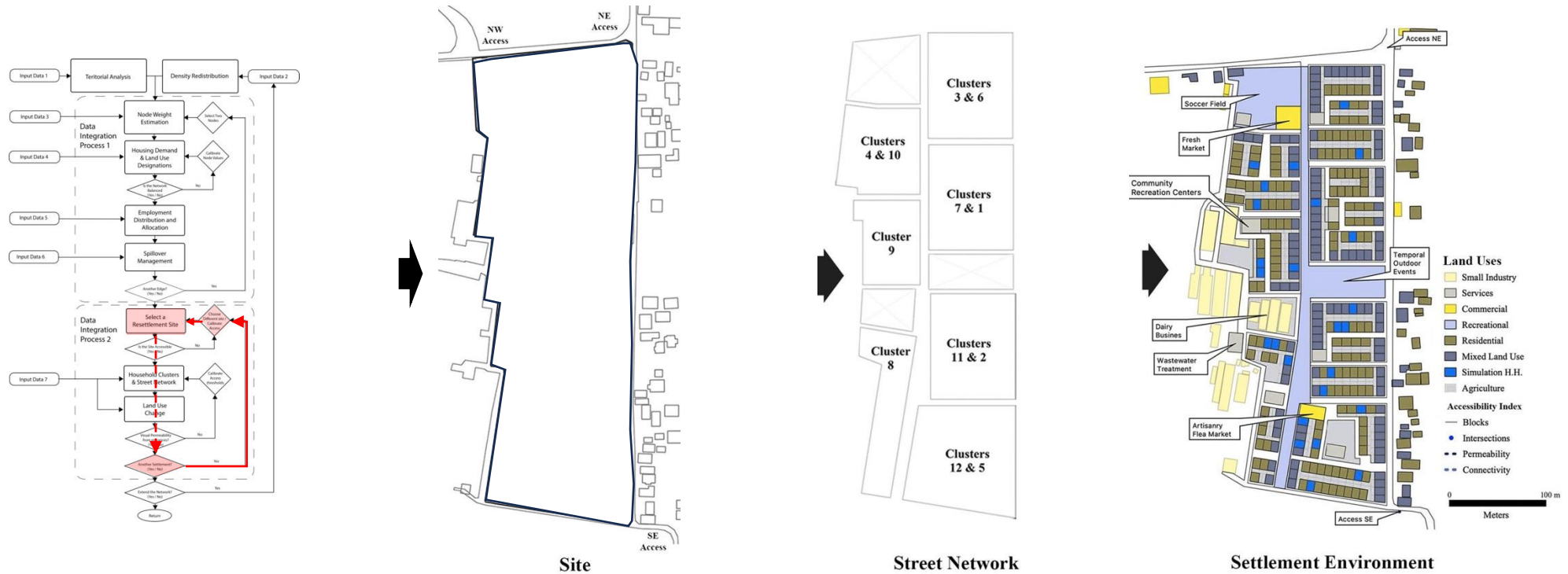
6 data inputs and models simulate urban development based on the allocation of primary land use activities in a road network.

Data integration process 2:

3 models and a few user inputs allow the simulation of a brand new settlement solution in a defined lot and evaluates land use change to locate households that will potentially venture a home-business in the area.



- The first data integration process relies on local employment multipliers to capture the demand for local services.
- Then, it allocates a firm or service to each node of a target area and estimates long-term employment as a result of attraction to allocated firms or services (Charpe, 2019; Moretti, 2010).
- New job vacancies are estimated, which attract people and increase housing demand. People tend to locate near jobs in nodes of the network, so a spillover management model allocates residential demand in sites sharing a side with roads of the network.



- The second data integration process allocates 301 synthetic households (proportional to the data sample) with common features in neighborhood clusters (Moeckel et al. 2003).
- Then, a settlement simulation component locates household clusters in a road network using the road features surveyed in GIS.
- Finally, agents were extracted from the synthetic dataset to pair households with mixed-use vacancies relative to their block.

1. Land-use treatment in nodes incited land-use conversion between the nodes.
2. People with higher education gained new jobs in nodes nearby increasing walking shares by 32%.
3. Increased transit incentivized home-business entrepreneurships (i.e., shops, small restaurants, hair salons, etc.) inside the settlement with 63% pertaining to farmers.
4. I found a significant probability of residential to mixed land-use change for middle-aged farmers and the elderly with risk-taking lifestyles and upper median incomes. This result helps to allocate particular households that will potentially launch home-businesses near roads.
5. Together with 30% travel utility savings, the network secured farming as the primary employment and mixed-use businesses as the secondary employment.

- The assumptions in Table 9 entail further research.
- Also, the data collected represents a portion of the ideal amount for microscopic simulations.

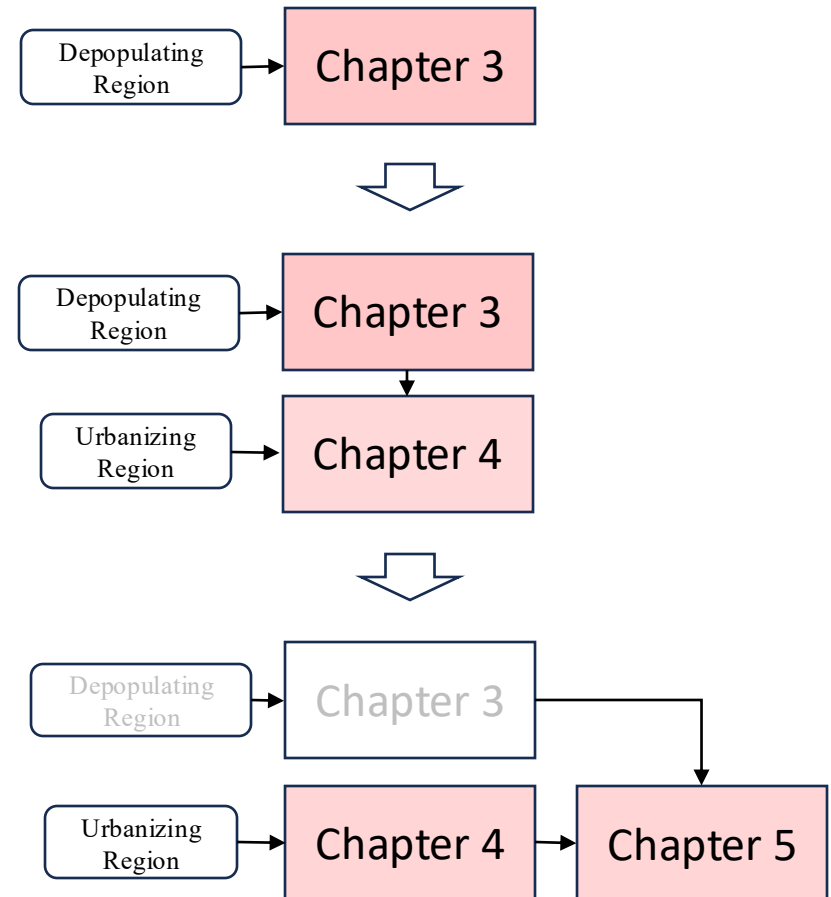
Table 9. OLUTM model assumptions

Module	Description	Theoretical Logic	Reference
Population Forecast	The government-recorded population growth rate is assumed to continue (the change in population over unit time period).	Population Growth Rates Cohort models	Statistical Census Bureau
Employment Distribution	The shift-share approach assumes that a change in national employment is unrelated to change in the local labor market.	Shift-Share Approach (National employment change replaced with district variance)	Charpe 2019
Employment Allocation	160 Non-tradable jobs were assumed to take hold of vacant lots in node 2, thus becoming HBEs.	Bid-Rent Theory of urban accessibility.	Alonso 2013
Location Choice	Relocating HHs place higher bids on lots with access to serviced areas.	Accessibility, Urban Economics, and Social Urban Theory.	Rhondi et al. 2018; Alonso 2013
Street Network	HBEs are driven by income, HH composition, and risk.	Prospect Theory	Kahneman and Tversky 1979
Land-Use Change	Livelihoods currently diversified reflect a choice of land-use change subject to a rise in people.	Relative Location in Bid-Rent Theory.	Rijanta 2008

Chapter 6. Conclusions

This study clarified relationships between population, land cover, and mobility associated to resettlement sites in areas affected by the 2011 Great East Japan Earthquake and the 2010 Merapi volcano eruptions, and demonstrates a model to evaluate resettlement sites.

1. A first method was proposed to explore population growth in a depopulating region (**Chapter 3**). Using regional characteristics obtained from the analysis, I proposed a unique method to help local governments find resettlement areas near urban structures likely to expedite urban growth.
2. A second method was proposed to explore land cover change in an urbanizing region (**Chapter 4**). This method estimated the maximum causal effects of resettlement to detect potential resettlement sites using the method in **Chapter 5**.
3. Based on **Chapter 4's** result, **Chapter 5** surveyed the local road network to find a resettlement site next to roads that sustain land use growth.
4. Then, **Chapter 5** evaluates household's mobility from the selected resettlement site and simulates a land use network to clarify whether rural labor can be sustained in an urbanizing region.



1. **Chapter 3** evaluated resettlement areas using post-disaster population data for Great East Japan Earthquake affected areas and found no significant change in population trends.
2. **Chapter 4** evaluated resettlement areas using land cover data for Merapi volcano affected areas and found small resettlement effects on land cover change with no significant trend alterations.
3. **Chapter 5** evaluated a settlement location after the Merapi eruption based on post-disaster mobility input data and found that settlement locations could be improved if livelihood options were located nearby.
4. Then, a simulation method was developed to deliver a predictive network of land use change and verify whether rural livelihoods were improved.
5. Based on the method's result, **Chapter 5** found that mobility and rural livelihoods were improved.
6. Limitations of this study include:
 - Time-consuming steps to complete the model.
 - Resettlement in depopulating regions will not have a significant impact on land cover change (based on the literature and results).
 - Results indicate that urbanizing peri-urban areas relative to Yogyakarta's wider region will experience similar results to those in this study. However, these results cannot be generalized to developed countries.
 - Although **Chapter 3** results are not meant for sub-regional cities in Miyagi and Iwate prefecture, resettlement near these cities could improve the overall exchange of goods and services between rural towns and cities promoting growth in these areas.
 - Six simulation model assumptions based on verified theoretical logic.

- The proposed model can rely on annual population census data for the past ten years and land cover change results from past resettlement cases in the ecozone of interest.
- Past resettlement cases can bring uncertainty.
- To handle this uncertainty, this study suggests that only local roads able to sustain more than the expected land use growth are surveyed as a safety measure to avoid land use dispersion.
- Another important source of data are mobility profiles, which can be collected from all residents relocating inside the local government's jurisdiction area using a survey questionnaire attached to this study.
- If the survey is conducted in a disaster risk area, then at least 8-10% of the target households in that area should answer the questionnaire to obtain statistically significant results.
- The model is for municipal local governments in need of identifying where future resettlement is likely to enable a successful recovery of the population and the affected area.
- It is also addressed to urban planners in private and public institutions who may act as consultants to the recovery planning efforts.
- Model application is suggested before a disaster happens to successfully address all components of the recovery process including pre-existing local issues, hence the title of this study: *“from a pre-disaster recovery planning perspective”*.

Appendix 2

User's Guide

Contents

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Figure 1 displays 16 heatmaps arranged in a 4x4 grid, illustrating the relationship between various combinations of industry and company size. The heatmaps are organized as follows:

- Top Row (Large Industry, Large Company):**
 - 1. **Large Industry, Large Company:** Heatmap with columns: CMO, CFO, COO, CEO, CHRO, CHGO, CHLO, CHCO. Rows: E&E, E&C, E&M, E&S.
 - 2. **Large Industry, Large Company:** Heatmap with columns: LHA, LHL, LLI, LNL, LPL. Rows: SH&S, SH&E, E&E, C&C, C&C.
 - 3. **Large Industry, Large Company:** Heatmap with columns: SH&S, SH&C, C&C, E&C, LHC, LHC.
 - 4. **Large Industry, Large Company:** Heatmap with columns: SH&S, LHA, LHL, LLI, LNL, LPL.
- Second Row (Large Industry, Small Company):**
 - 5. **Large Industry, Small Company:** Heatmap with columns: CMO, CFO, COO, CEO, CHRO, CHGO, CHLO, CHCO. Rows: E&E, E&C, E&M, E&S.
 - 6. **Large Industry, Small Company:** Heatmap with columns: LHA, LHL, LLI, LNL, LPL. Rows: SH&S, SH&E, E&E, C&C, C&C.
 - 7. **Large Industry, Small Company:** Heatmap with columns: SH&S, SH&C, C&C, E&C, LHC, LHC.
 - 8. **Large Industry, Small Company:** Heatmap with columns: SH&S, LHA, LHL, LLI, LNL, LPL.
- Third Row (Small Industry, Large Company):**
 - 9. **Small Industry, Large Company:** Heatmap with columns: CMO, CFO, COO, CEO, CHRO, CHGO, CHLO, CHCO. Rows: E&E, E&C, E&M, E&S.
 - 10. **Small Industry, Large Company:** Heatmap with columns: LHA, LHL, LLI, LNL, LPL. Rows: SH&S, SH&E, E&E, C&C, C&C.
 - 11. **Small Industry, Large Company:** Heatmap with columns: SH&S, SH&C, C&C, E&C, LHC, LHC.
 - 12. **Small Industry, Large Company:** Heatmap with columns: SH&S, LHA, LHL, LLI, LNL, LPL.
- Bottom Row (Small Industry, Small Company):**
 - 13. **Small Industry, Small Company:** Heatmap with columns: CMO, CFO, COO, CEO, CHRO, CHGO, CHLO, CHCO. Rows: E&E, E&C, E&M, E&S.
 - 14. **Small Industry, Small Company:** Heatmap with columns: LHA, LHL, LLI, LNL, LPL. Rows: SH&S, SH&E, E&E, C&C, C&C.
 - 15. **Small Industry, Small Company:** Heatmap with columns: SH&S, SH&C, C&C, E&C, LHC, LHC.
 - 16. **Small Industry, Small Company:** Heatmap with columns: SH&S, LHA, LHL, LLI, LNL, LPL.

Each heatmap is labeled with its corresponding industry and company size combination. The color scale indicates the strength of the relationship, with red representing low and green representing high.

*Each household group is defined by a square of four alternatives within each colored land use block. These admit compatible singles whose land use match the ones already defined within a group.

1.1.4. Classifying survey data (Lifestyle Segmentation Data)

1.1.4^a. From Lifestyle segmentation

Each person has been assigned with a lifestyle segment according to the initial survey questionnaire using VALS™ or "Lifestyles" by Arellano:

VALS (Developed Countries)

1. Innovators (I) 2. Thinkers (T) 3. Believers (B) 4. Achievers (A) 5. Strivers (S)
6. Experiencers (E) 7. Makers (M) 8. Survivors (S2)

Arellano Marketing (Developing Countries)

1. Sophisticated (S) 2. Progresist (P) 3. Modern (M) 4. Formalist (F) 5. Conservative (C)
6. Austere (A)

1.1.4^b. The number of "states" is a function of the number of adult occupants per household.

In a community-based neighborhood unit, a cluster of houses that meet each other's needs and engage in proactive activities consists in no more than 30 houses for a 100 x 100 meter neighborhood area (丁目).

Results from the survey in Appendix 1 can provide household (HH) compositions:
 (30 houses $\times$ 0.6; 1-adult HH's) + 2*(30 $\times$ 0.3; 2-adult HH's) + 3*(30 $\times$ 0.1; +3-adult HH's)
 =100%, 45 Adults in total
 30 HH's, 10,000 sqm, 180 sqm / HH, 5,400 sqm built area, 4,600 sqm unbuilt area (46%)

VALS:
1 Adult HH: $30 \times 0.6 = 18$ H.H

1 Adult HH: $30 \pm 0.3 = 9$ H.H.

for new development during the next ten years (The enclosure ratios used are limited to the surveyed areas). *Note: Minimum housing area required is 90m².*

5.1.8. If a selected site is located within the influence area of a node, apply ratios to periphery roads.

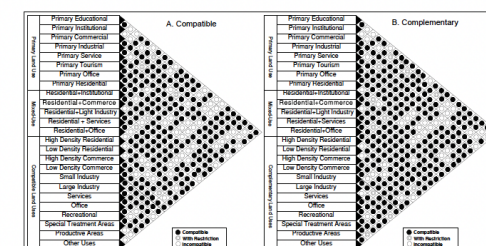
6. Land Use Allocations

6.1. Allocate areas for new households destined to live in the AOI. Each household has 1, 2, or 3 adults per HH. Therefore, assign local standard areas for each type of HH. We used 70m² (1-adult), 100 m² (2-adult), and 120 m² (+3-adult). The remaining area is used for non-residential development.

6.2. Compatible Land uses: Activities that work together or in synergy by need or incentive.

6.3. Road hierarchy is used to classify land sharing a side with roads. Subsequently, existing structural activities categorize each node of the AOI with land use categories: Housing, Commerce, Small Industry, Large Industry, Office, Service, or multi-purpose. Structural activities are services, firms, or other land use that attracts people from outside the district (inter-district nodes) or outside the prefecture (metropolitan nodes). Using this knowledge, new structural activities are required in nodes where district and inter-district structural activities do not exist.

6.4. Using the remaining area forecasted for development, assign compatible land uses to the structural activities in each nodal influence area. Use matrix 1 below.



Matrix 1. Compatible and complementary land use allocations

6.5. Complementary Land Uses

6.5.1. Complementary Land Uses: Activities that add value to neighboring activities.

6.5.2. Classify road hierarchies and apply enclosure ratios for housing in local roads and collector roads when site is not found within the nodal's area of influence.

6.5.3. Include neighborhood activities from Matrix 2.

Publications

[Peer-reviewed article]

Garcia-Fry, M., Murao, O., Bachri, S., Moya, L.A., 2022. Land-Use microsimulation model for livelihood diversification after the 2010 Merapi volcano eruptions. Transp. Res. Part D: Transp. Environ. 104, 103189. <https://doi.org/10.1016/j.trd.2022.103189>

[Peer-reviewed abstract]

Garcia-Fry, M. and Murao, O., 2022. Retrieving small area population dynamic trends after the 2011 Great East Japan Earthquake. AI-WEST DR 2022, Priority 4: Enhancing disaster preparedness for effective response and to “Build Back Better” in recovery, rehabilitation, and reconstruction. 2022.09.

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9th Joint Student Online Seminar on Civil Infrastructures. International Center for Urban Safety Engineering. 12/08/2020–12/10/2020. https://icus.iis.u-tokyo.ac.jp/wp/wp-content/uploads/2021/04/89_9th-joint-student-seminar2020.pdf



-Garcia-Fry, Martin (2021), “OLUTM Model Dataset”, Mendeley Data, V3, doi:
<http://www.doi.org/10.17632/t9p23k3pyn.3>

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